

RFP NO: OT/EC/3/56/2026-27/2193

**NAME OF WORK: PROVN OF 01 x COOK HOUSE AND DINING HALL IN
THE AOR OF 56 INF DIV**

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COMMANDING OFFICER, 110 ENGINEER REGIMENT C/O 99 APO REQUEST FOR PROPOSAL (RFP) INVITATION OF ONLINE BIDS FOR “PROVN OF 01 x COOK HOUSE AND DINING HALL IN THE AOR OF 56 INF DIV” NO OT/EC/3/56/2026-27/2193

1. Tender documents for the above-mentioned work are uploaded herewith on www.gem.gov.in. You are authorised to participate in this competitive tendering by uploading tender in two covers i.e. **one is technical bid/applications and the second is price /financial bid**. However the President of India does not bind himself to accept the lowest or any tender.
2. All critical dates i.e. publishing dates, downloading dates and bid submission dates shall be referred as per the NIT details mentioned and updated in the website.
3. Tender documents for the above work have been uploaded on website www.gem.gov.in.
4. Instructions for completion of the tender documents are also given which should be complied with. Contractor's particular attention is invited to the same, which deal with e-tendering process and some new eligibility criteria which is now being introduced in the Deptt.
5. Tenderers to note that NO OFFLINE TENDER SHALL BE ENTERTAINED. The tender documents are available online on website "www.gem.gov.in". The interested tenderers/firms must have Digital Signature Certificate (DSC) functional and duly registered on website www.gem.gov.in as bidder.
6. Tenderer shall note that information technology Act, cyber Act along with other Act as relevant shall be applicable.
7. Instruction for participating & filling of tender and all documents referred along with this letter shall be forming part of tender documents and shall be complied with.
8. The address and contact numbers for seeking any clarifications regarding this RFP are given below :-

- | | | |
|---|---|---|
| 8.1. Queries to be addressed to | : | Commanding Officer,
110 Engineer Regiment. |
| 8.2. Postal address for sending the Bids | : | Commanding Officer,
110 Engineer Regiment,
C/o 99 APO
Pin 914110 |
| 8.3. Name/Designation of the contact person : | | Col Rishi Katiyar,
CO, 110 Engr Regt |
| 8.4. Official Contact Number | : | - |
| 8.5. E-mail Id of contact personnel | : | risyar.985@gov.in |

Sd/- x- xx xxx
(Rishi Katiyar)
Col
Commanding Officer

**INSTRUCTIONS FOR SUBMISSION OF BID/ TENDER DOCUMENTS
TO BE COMPLIED BY THE TENDERER**

1. **Earnest Money Deposit** : All eligible Contractors are required to submit Earnest Money Deposit ~~but have not executed the Standing Security Bond shall submit Earnest Money Deposit~~ as per details in NIT either in the form of Call Receipt Deposit or FDR from a Schedule Bank in favour of Commanding Officer, 110 Engr Regt. Non-submission of earnest money will render the tender as non-bonafide and consequently rejected.
2. **Performance Security Deposit**: The performance security shall be in favour of Accepting Officer in any of the forms to include Bank Guarantee (Issued by Nationalised /Scheduled Bank with modifications with respect to Performance Security in place of Security Deposit), Govt Security for an amount equivalent to 5% of the contract sum before Supply Order. In case a fixed deposit receipt of any bank is furnished by the contractor to the Government as part of the Performance Guarantee and the bank is unable to make payment against the said fixed deposit receipt the loss caused thereby shall fall on the contractor and the contractor shall forthwith on demand furnish additional security to the Government to make good the deficit. Noncompliance of this provision shall constitute sufficient grounds for cancellation of awarded contract and forfeiture of Earnest Money. In case contractor fails to complete the work under the contract then the Performance Security deposit will be deposited in the govt treasury through MRO.
3. **General Instructions for Compliance**
 - 3.1. The tender will only be accepted in the electronic form. All bids to be submitted on website "www.gem.gov.in" portal. No offline submission of tender is acceptable. The tendering is based on 2 cover system. The tenderer shall submit the documents in requisite format as stipulated in NIT & tender documents and as mentioned herein after.
 - 3.2. Cover wise contents & documents are given herein after. All the enclosures/attachments to be uploaded by bidder for tech bid shall be in pdf format. The bill of quantities (BOQ)/price Bid shall be uploaded under fin bid and shall be in Excel sheet (.xls) format. All the documents must be signed by the bidder before submission. The list of documents & content are given as under:-
 - 3.2.1. **Tech Bid.** The documents/details must be scanned & converted in pdf form before uploading to include all the documents as per list given below. **(All the requisite documents must have quoted tender/bid ID/No and date on Firm letter Head wherever applicable)**
 - 3.2.2. **Eligibility Criteria:-**
 - 3.2.2.1 **Bidder Turnover:-** The minimum average annual financial turnover of the bidder during the last three years, ending on 31st March of the previous financial year, should be as more than 200% on total tender value. Documentary evidence in the form of certified audited balance sheets of relevant periods or a certificate from the Chartered Accountant/Cost Accountant indicating the turnover details for the relevant period shall be uploaded with the bid. In case the date of constitution incorporation of the bidder is less than 3 year-old, the average turnover in respect of the the completed financial years after the date of constitution shall be taken into account for this criteria.

3.2.3. Experience Criteria :-

3.2.3.1 The bidder should have regularly executed same or similar category work at any Central/State Govt Organization/PSU Public Listed Company in last 3 years before the bid opening date. Copies of relevant contracts to be submitted is as follows:-

3.2.3.1.1. Three works costing not less than **40% of estimated cost**
OR

3.2.3.1.2. Two works costing not less than **50% of estimated cost**
OR

3.2.3.1.3. One work costing not less than **80% of estimated cost**

3.2.4. Quality testing certificate of machinery and equipment not later than 2 year be uploaded with technical bid.

3.2.5. Self declaration certificate that the site has been recced and there are no issues with respect to site for construction of the said job.

3.2.6. **Compliance of Tender Documents.** The contractor should read all pages of tender documents and upload the same with bid duly signed and stamped.

3.2.7. **Past Performance.** Bidder should upload the Past performance certificate issued by respective buyer organization with SATISFACTORY and above remarks in atleast last one of the last two financial year for the same/ similar work on Supply work not less than 50% of the tender value. The works should be carried out only within Manipur, Arunachal Pradesh, Tripura and Assam in the AOR of HQ Eastern Comd region.

3.2.8. Documents to be submitted under details of the firm:-

3.2.8.1. Application for bid participation in Firm's letter head.

3.2.8.2. Scanned copy of DD towards Cost of tender (original copy of EMD deposit to be handed over to CO, 110 Engineer Regiment before bid opening date).

3.2.8.3. Scanned copy of GST Registration.

3.2.8.4 Any other documents which the bidder feels relevant or mentioned in NIT.

3.2.8.5. Scanned copy of EMD deposit in form of FDR/PGB (original copy of EMD deposit to be handed over to CO, 110 Engr Regt before bid opening date).

3.2.8.6. Affidavit that the applicant firm is not black listed or banned by any Govt.

3.2.8.7. Affidavit that no Govt recovery is due against the firm.

3.2.8.8. Copy of valid passport or police verifications of each of the director/partner or proprietor of company/firm.

- 3.2.8.9. Supply order/Work order details during last 03 years as per format given and other details to prove eligibility as per Manual of Contract 2007 (Available on internet). To include copies of Supply/Work order issued by any Govt department of works of similar nature/more than similar amount of work.
- 3.2.8.10 Copy of Income Tax Return for last 03 years OR Form 26 AS for last 03 yrs.
- 3.2.8.11. Turnover certificate for last 03 years duly signed by CA.
- 3.2.8.12. List of work in hand under progress as on date of submission of bid in firm letter head (Work order/Supply order copy).
- 3.2.8.13. Bid Security declaration certificate.
- 3.2.8.14. Non-disclosure certificate.
- 3.2.8.15. Store delivery assurance certificate.
- 3.2.8.16. Adhar, Pan, firm GSTIN, NEFT Mandate form & cancelled cheque.
- 3.2.8.17. Certified copy of PAN No. /TIN No./GST No of proprietor / partners / firm.
- 3.2.8.18. Indemnity Bond on non-judicial stamp paper of requisite value duly attested by Notary (on format given below)
- 3.2.8.19. Contractor should have worked in **Manipur/Arunachal Pradesh/Assam/Tripura** location for last 03 financial year.
- 3.2.8.20. **Solvency Certificate.** The bidder has to produce solvency certificate along with the bid. The solvency amount to be more than the contract value (Format att as **Appx 'A'**).
- 3.2.8.21 User manual to be printed and supplied in hard & soft copy along with the sample structure.
- 3.2.8.22. **Termination/ Cancellation of Supply Order.** In the event of slow progress/ work not started against the Supply Order offered to you for those items shall be terminated/cancelled instantly with respect to following work progress: -
- 3.2.8.22.1. Work not started within 01 week (07 days) after issuing work order.
- 3.2.8.22.2. Overall 50% of work has not been done within 01 months (30 days) after issuing work order.
- 3.2.8.22.3. Overall 100% of work has not been done within 02 Months (60 days) after issuing work order.

NOTE: - In the event of breach of contract on any account or no supply order of materials or delay in supply of work, Commanding Officer, 110 Engineer Regiment shall have the **right to cancel the supply order** and purchase the stores at your **RISK AND COST**. Additional expenditure incurred shall be recovered from your firm. Undertaking certificate in this regard **ON FIRM LETTER HEAD** should be

submitted during technical bid. Format is as under: -

UNDERTAKING CERTIFICATE FOR TERMINATION/CANCELLATION OF WORK ORDER

It is certified that this firm M/s _____ has agreed the following terms and conditions against work progress on supply of the said job which fails Commanding Officer, 110 Engineer Regiment shall have the **right to cancel the supply order** and purchase the stores at our firm's **RISK AND COST**. Additional expenditure incurred shall be recovered from our firm: -

1. Work not started within 01 week (07 days) after issuing work order.
2. Overall 50% of work has not been done within 01 month (30 days) after issuing work order.
3. Overall 100% of work has not been done within 02 Months (60 days) after issuing work order.

Place:

(Name of the proprietor)
(With seal and signature)

Date:

Appendix –A**FORM OF SOLVENCY CERTIFICATE FROM A SCHEDULED BANK**

This is certified to the best of our knowledge and information M/s/Shri/Smt. _____ having address _____, a customer of our bank are/ is respectable and can be considered solvent/financially sound for any engagement upto Rs. _____ (Rupees _____). This certificate is issued without any guarantee or responsibility on the bank or any of the officers.

Signature of Bank Manager Place:
Name, No. & Seal of Bank Date:

Note:- In case of partnership firm, certificate to include names of all partners as recorded with the bank.

ADDITIONAL TERMS AND CONDITIONS: - (On Bidder Letter Head) to be uploaded with bid):-

1. **Clause:-** The Vendor participating in the bidding must have certified a Fabrication Unit (Proof / certificate should upload alongwith technical bid)
2. ISO certification ISO 9001 (Quality and safety), 9000 (Quality Management).
3. Lab report from authorized source for PUF panel.
4. **NON-DISCLOSURE DECLARATION CERTIFICATE**
 - 4.1. I M/s. NAME OF FIRM hereby declare that I shall not disclose the contractor any provision, specification, plan design, pattern, sample or information thereof to any third party during and after expiry of Contract.
 - 4.2. If defying the norms as per the Contract a legal action may be taken againstme as per the existing Rules.
5. **CONFIDENTIALITY CERTIFICATE** It is certified that the Company or any representative of the Company or Agents authorized by the Company will notdisclose any information gained by them or their representatives or Agents, while interacting with the persons of the 110 Engineer Regiment, C/o 99 APO or any other office of the Headquarters 3 Corps, c/o 99 APO or any documents prepared in connection with the project or any documents received by them, or any study carried out by them, directly or indirectly to any person or Company or Institution or Press. Anyviolation therein may be considered as a violation of Indian Official Secret Act 1923.
6. **STORE DELIVERY ASSURANCE CERTIFICATE.** I hereby confirm that complete store is available for delivery within delivery period at FOR loc. No extensionwhatsoever will be requested w.r.t the delivery period.
7. OEM certificate of the firm as per specifications mentioned in specificationdocument uploaded with the bid.
8. Bidder's offer is liable to be rejected if they don't upload any of the certificates /documents sought in the Bid document, ATC and Corrigendum if any.
9. Undertaking Certificate for the rates quoted will remain valid for six months fromthe date of opening of quotation for repeat/option clause.
10. Bidder should have a **OEM auth/dealership certificate for cement/ OPC.**
11. **Sample Clause:-** All the bidders must attend pre-bid meeting and produce the sample as per technical specification of item prior to opening of the technical bid. Only the bidders with approved sample shall be considered for further technical and financial bidding. Bidders who fail to produce the sample and sellers whose sample gets rejected by the Board of Officers shall be rejected at technical stage.

INDEMNITY BOND

This bond of indemnity is executed on this day _ by M/s Shri having registered office at To indemnify the Govt of India of the following:-

1. We undertake to pay the Government of India any damages that may be found to be recoverable on order of our contracts or failure to execute the contract.
2. We undertake that in case Government is put to pay any loss or disadvantage in execution of contract, the firm shall indemnify the government for all loss or disadvantage.
3. This bond of indemnity is executed by M/s_ and have signed before me. Whatever stated above is true and correct to the best of my knowledge and belief.

Signature of Contractor

Notary

3.1. Notarised affidavit that there is no relationship with any Gazetted/Commissioned Officers and JEs in Department/Corps of Engineers/Ministry of Defence. In case of relation the same shall be disclosed with details.

3.2. An affidavit affirming that all the information and details furnished are true and the bidder may be liable for suitable administrative and legal action in case the information and details are found to be false/ incorrect.

Note:- All affidavits must be submitted duly notarized and shall be on Nonjudicial stamp paper of requisite value.

FIN BID : Bill of quantities (price/financial Bid) shall be uploaded by bidder duly digitally signed in Excel sheet (.xls) format within the duration (time) allowed for bid submission. This shall be uploaded by contractor along with documents of cover-I. The price Bid/BOQ of those contractors who are considered technically qualified as per documents in cover-I only shall be opened.

1. All bids and documents will be digitally signed by the contractor.
2. The tender documents along with all concerned documents shall be published and made available in public domain (on website www.defproc.gov.in). The time and date fixed under "Critical Dates" heading shall be complied by tenderers & Deptt respectively as applicable.
3. The tenderer shall login through by creating his login Account on above said website to participate in tendering. The bidder shall download complete tender documents & go through it thoroughly and quote his rates in BOQ (cover 2 enclosures) after downloading the same. The Bidder has to enter name of his firm/Company & the Lump Sum in relevant column & location and submit/ upload his bid after digitally signing the same. The bidder shall note that he will not modify or make any change in any of the documents uploaded by the deptt. In case any such modification/tempering is noticed at any stage, administrative and other action as suitable as per law shall be taken. Contractor shall note that after the date and time fixed for submission of bids no correction/revision can be made to any document. Contractor shall employ Indian national only after verifying their antecedents and credentials. Attention is drawn to condition No 24 & 25 of IAFW- 2249. Subletting of work/contract in any form is not allowed. In this connection Condition No 18 & 53 of IAFW-2249 shall be referred. For submission of Bid (Technical & price both), the contractor shall ensure

timely submission of all documents as per NIT & Tender stipulation.

4. In the event of lowest tenderer revoking his offer or revising his rates upwards/offering voluntary reduction, after opening of the Technical / Price Bid, the Earnest Money deposited by him shall be forfeited. Reduction offered by the tenderer on freak high rates referred to the tenderer for review shall not be treated as voluntary reduction.

5. Description of items of work mentioned in "Schedule of work – BOQ" shall be read in conjunction with notes given under Sch 'A' (Schedule of works - BOQ). The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. The tenderer /bidder shall quote his rate on the BOQ file only. No alteration to the format will be accepted; else the bid will be disqualified and summarily rejected. Modification to the rates quoted in the BOQ (excel sheet) possible before the bid closing time and date.

6. The tenderer is expected to be fully conversant with the technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literatures on the subject are widely available which the tenderer may make use of. The tenderer is responsible for supply of all the stores/materials and construction of the shelter as per the specification specified in the tender documents. In case of any ambiguity on the design/specification, the minimum scope as per the provisions given in IS code/ MES SSR-2020 (Part-II) to be provided as per normal engineering practices.

7. The tenderers attention is drawn to special conditions of the tender regarding preparation of the detailed network and time schedules as for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer in using the technique will be taken as his technical insufficiency and will affect his class of enlistment and future prospects for receiving tenders for the works.

8. The department may issue amendments/corrigendum to the tender documents before date & time fixed for start of bid submission. The tenderer is required to read tender documents in conjunction with the amendments if any, issued by the department. Only the errata/amendments issued by the department shall hold good.

Detail of work of Provn

Provn of 01 X Cook House And Dining Hall in the AoR of 56 Inf Div:-

Ser No	Job No	Description	Location of work	Paying Authority	Qty of Assets
8.1.	OT/EC/3/56/2026-27/2193	Provn of 01 X Cook House And Dining Hall in the AoR of 56 Inf Div.	Yinkiong	Commanding Officer, 110 Engr Regt	01

These instructions shall form part of the contract documents.

Sd/- x xx xxx
(Rishi Katiyar)
Col
Commanding Officer

Sig of Contractor _____

**110 ENGINEER REGIMENT
NOTICE OF TENDER**

1. A tender is invited from the contractor's meeting eligibility criteria for the work as mentioned in para 3 of page 3 of instruction for submission of bid documents.
2. The work is estimated to cost as indicated in Appendix 'A' to Notice of Tender. This estimate however is not a guaranteed and it is merely given as rough guide and if the work costs more or less, the tenderer will have no claim on this account. The tender shall be based as mentioned in Appendix 'A' to Notice of Tender.
3. The work is to be completed within the period as indicated in aforesaid Appendix 'A' to Notice of Tender in accordance with the phasing, if any, indicated in the tender from the date of handing over site.
4. One tenderer/bidder or one firm of the contractor shall participate in not more than one online submission of tender. Under no circumstances will a father and his son(s) or other close relations, who have business dealing with one another be allowed to tender for the same contract as separate competitors. Breach of this contract condition will render the tender of both parties liable to rejection.
5. The Commanding Officer, 110 Engineer Regiment, c/o 99 APO will be the Accepting Officer, hereinafter referred to as such for the purpose of this contract.
6. The tenderer/bidder are required to submit all the documents as given in para 3.2(a) of instruction for submission of bid documents. In case of any deficiency in the documents the same will be intimated to the bidder through document short fall on the portal and bidder will have opportunity to submit the same within prescribed period.
7. Complete status of bids Tech bid & Fin bid and its opening etc will be updated on website immediately of the happening of the event or change in status. Submission of documents under tech bid & price bid under fin bid does not guarantee the acceptance thereof. The price bid of all bidders shall be opened subject to the condition that the bidder qualifies the laid down eligibility criteria and conditions of the tender.
8. Tender forms and conditions of contract and other necessary documents will be uploaded in Tech bid & Fin bid as per date given in aforesaid Appendix 'A' to Notice of Tender.
9. The Commanding Officer, 110 Engineer Regiment will return the earnest money applicable to all unsuccessful tenderers by endorsing an authority on the deposit receipt for its refund. All tenderers must be very careful to upload a bonafide tender, and its related documents failing which the Accepting Officer may at his absolute discretion, reserve the right of forfeiting a portion of Earnest Money, Security Deposit or Security Bond amount (not exceeding amount of cost of tender) deposited by the tenderer. A bonafide tender must satisfy each & every condition laid down in this tender.
10. Contractor quoted rates shall be deemed to include amendments/ corrigendum's to tender documents published till last date & time of closing of bid submission. Contractor shall regularly check the corrigendum's published to keep them upto date.
11. Contractor shall quote his rate on the latest BOQ published online with reference to the tender. Bids quoted on previous BOQ shall be summarily rejected and not considered as Bonafide and suitable necessary action may be initiated against the contractor.

12. The tenderers are advised to visit the site by making prior appointment with the Engineer Regiment. A tenderer shall be deemed to have full knowledge of all related documents, samples, site etc. whether he has inspected site or not. The tenderers can submit the tender (Tech bid & Fin bid) up to the last date & time of submission of bid and the bid will be opened online by the opening officers on the date & time mentioned / stipulated.
13. Copies of the drawings and other documents pertaining to the work and sample of material and stores to be supplied by the contractor will be kept open for inspection of the tenderers in the office of Commanding Officer, 110 Engineer Regiment as mentioned in the aforesaid Appendix 'A' during working hours.
14. The Accepting Officer reserves the right to accept a tender submitted by a public sector undertaking giving a purchase preference over other tenders(s) which may be lowest, as are admissible under the Govt policy. No claim for any compensation or otherwise shall be admissible from such tenderers whose price bid (fin bid) may be rejected on account of said policy.
15. Commanding Officer, 110 Engineer Regiment will upload status of tender on the date and time indicated in the aforesaid Appendix 'A' to this Notice of Tender.
16. Any tenderer, who puts any alterations to any of the conditions, laid down or puts any other condition of any description whatsoever in his bid submitted is liable to be rejected.
17. The submission of tender by a tenderer implies that he has read the conditions of the tender and has made himself aware of the scope and specifications of work to be done, local conditions and other factors bearing on the execution of the work. This shall also include all the taxes, GST, entry tax, vitality tax of forest department etc.
18. The Accepting Officer does not bind himself to accept the lowest tender or any tender or to give any reason thereof.
19. Subletting of work is not permitted.
20. This Notice of tender shall form part of the contract.
21. Scanned copy of DD with tender/bid enlistment details/documents shall be uploaded as Tech bid of the tender/bid one-tendering portal. DD is refundable in case T bid is not accepted resulting in non-opening of 'Q bid. The applicant contractor shall bear the cost of bank charges for procuring and encasing the DD and shall not have any claim from the Government whatsoever on this account.
22. Tender form and conditions of contract and other necessary document shall be available on www.gem.gov.in website for download and shall form part of contract agreement in case the tender/bid is accepted.

Sd/- x xx xxx
(Rishi Katiyar)
Col
Commanding Officer

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

1.	Name of work	Provn of 01 x Cook House And Dining Hall in the AoR of 56 Inf Div
2.	Estimated Cost of Work	-
3.	Cost of tender documents	Not Required
4.	Period of completion	60 days from the date of Supply Order
5.	Website/portal address	https://gem.gov.in
6.	Type of Contract	The tender shall be based on Lump Sum contract for carrying out complete supply of stores and execution of the shelters at site (inclusive of all the cost required to complete the work).
7.	Information & Details:	
	7.1. Bid submission start date	As per critical dates to be published online
	7.2. Last date of bid submission	As per critical dates to be published online
	7.3. Date of bid opening	As per critical dates to be published online
8.	Eligibility Criteria	As per para 3 of page 3 of instruction for submission of bid documents.
9.	Amount for bid security/earnest money deposit	Eligible Bidders are required to submit Earnest Money Deposit (EMD) @2% of bid value (if applicable) along with their bids. FDR/Bank Guarantee on account of Earnest Money for contracts being concluded at this unit HQ will be made in the name of Commanding Officer, 110 Engineer Regiment C/o 99 APO only the earnest money of the lowest tenderer will be retained till submission of performance Bank Guarantee of 5% on the total cost value of the award of contract and within 15 days from the date of issue of supply order and refund of performance security released after expiry of defect liability period. Defect liability of this work shall be 24 (Twenty-Four) Calendar Months from the certify date of completion of work.
10.	Tender issuing and Accepting Officer	Commanding Officer, 110 Engineer Regiment
11.	Executive Agency	110 Engineer Regiment

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (Contd...)

Critical Dates. The critical dates with respect to the Tender ref OT/EC/3/56/2026-27/2193.

Ser No	Item	Date	Time
1.	Published Date	Refer critical dates uploaded on www.gem.gov.in	
2.	Bid Submission End Date		
3.	Bid Opening Date		

Notes :-

- Only Indian firms are eligible to participate in the bidding.
- Tender not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in tech bid shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.
- After opening of tech bid and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of Shortfall in DPP shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid shall not be opened and contractor shall not have any claim on the same.
- Invitation for e-tender docs not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in para 3 of page 3 of instruction for submission of bid documents. Tenderer/bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website.
- In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (ie he has quoted on pre revised BOQ). Such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.
- Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a willful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.
- Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

8. **EPF Code & UAN:** All contractors shall mandatorily be in possession of the EPF Code Number and UAN of employees working under their establishment under EPF & MP Act 1952. So far this tender is concern, before release of Supply Order/Work Order the contractor shall submit EPF Code Number to the Engineer Regiment.

9. **Defect Liability Period:** The Defect Liability Period in respect of this tender shall be Twenty Four (24) months. In case contractor will not rectify any defects during defect liability period, recovery of the same shall be effected from the Performance Security.

10. **Refund of Performance Security:** The Performance Security Deposit shall be refunded to contractor after expiration of Defect Liability Period. Recovery if outstanding against the contractor shall be effected from the Performance Security as stipulated under Condition 67 of IAFW-2249 (as amended).

Sd/- x xx xxx
(Rishi Katiyar)
Col
Commanding Officer

GENERAL CONDITIONS

1. **PERIOD OF COMPLETION.** The entire work shall be completed within **60 days** from the date of supply order.
2. The work under this contract shall be executed in restricted area where working hours and **routes are restricted by the authorities controlling the area for security purpose**. This being vital factor regarding loss of man-hours etc. shall be considered by the contractor while quoting his rates for this work. Govt. shall not entertain any claims whatsoever on this account at subsequent date.
3. Probable distributions of various items are indicated on drawing these are tentative and may be varied where necessary at the discretion of Commanding Officer. The contractor shall not be entitled for any claim on account of such varied quantities, alignment or change in re- quantities, no price adjustment shall be done on account of final approved layout within the site plan area.
4. Unit rate quoted by the contractor against various items shall, inter alia, be deemed to include for any minor details of construction which are obviously and fairly intended and which may not have been specifically included in the tender documents but which are essential for the execution and completion of works. In case of difference of opinion between the contractor and the EPO as to whether or not, any item of work constitutes 'Minor details of construction', the decision of the Commanding Officer, 110 Engineer Regiment (Accepting Officer) shall be final, conclusive and binding.
5. Unless otherwise specified, unit rate against each item of work shall be deemed to be inclusive of materials and labour, all plants, equipment's and tools including testing for acceptance criteria complete.
6. The defects liability period mentioned in offer competition of Two year shall commence **from the date of completion of the entire work** under the contract. Date of completion shall be clearly indicated vide a letter stating the same by the EPO, on behalf of Govt. of India.

General Note: -

1. **e-Performance Bank Guarantee.** Performance security deposit in favour of Accepting Officer @ 5% of the amount of this Supply Order amount shall be deposited by the Contractor in form of e-Performance Bank Guarantee/BGB issued by a public sector bank or a private bank authorized to conduct government business within 15 days from the date of issue of supply order as per prescribed format. Vendor shall ensure that PBGs to be submitted by him should preferably be sent to Commanding Officer, 110 Engineer Regiment directly by Issuing Bank by registered post (AD). PBG should remain valid for a period of 24 Months beyond the successful completion of all obligations under this supply order. Contractor must revalidate Bank Guarantee, in case of grant of extension of Supply and period (as the case may be).
2. **Payment Instruction (e -payment) :-**
 - 2.1. Payment will be made in part or full, subject to receipt and inspection of stores/ constr of asset and satisfaction of the Board of Officers at designated location.
 - 2.2. Prescribed pre-stamped, pre-receipted bills in triplicate will be submitted to the paying authority after fulfilling (a) above.

2.3. Original copy of supply order will be attached with the bill including necessary test certificates/instruction schedule.

2.4. **Payment Schedule.** On completion of following stages, payments will be released by the Engineer Regiment on successful ground verification.

2.4.1. On delivery at site : 100%

2.4.2. Paying Authority: Commanding Officer
110 Engineer Regiment
PIN – 914110
c/o 99 APO

3. **Taxes.** Firm will ensure that all the taxes are paid for stores supplied against this supply order and payment, as applicable, by the supplier and no additional cost whatsoever will be borne by the consignee.

4. The supplier will quote his GST Registration Numbers and endorse the following certificate on the bill:-

“Certified that the goods on which GST has been charged, have not been exempted under the tax Act or the rules made there under. The charges levied on account of GST on the goods supplied are correct under the provision of the Act or the rules made there under”.

5. Where excise duty is claimed, the following certificate should be attached to the bill.

"Certified that the excise duty has been paid, have not been exempted, as noted under the Excise Duty Act. The excise duty charge on the bills is of what has actually been paid".

6. The contractor is responsible for design, execution and the entire scope of work. Therefore the contractor to visit site and assess the site/foundation condition. Nothing additional will be payable on any account for the complete scope of work.

7. In case any item of work left out and if found necessary to be undertaken for efficient use of the facility must stringent of the requirement to be provided.

8. **Tender Inviting Authority:** Commanding Officer, 110 Engineer Regiment

Name of Work/ Job:- PROVN OF 01 X COOK HOUSE AND DINING HALL IN THE AOR OF 56 INF DIV

Name of Contractor :- M/s

Please refer your offer in response to this unit Contract No **OT/EC/3/56/2026-27/2193** and rates quoted thereon by you.

9. The rates quoted by your firm against our above-mentioned enquiry for stores as per schedule below have been accepted. You are requested to Supply of Work to 110 Engineer Regiment as per the terms & conditions specified at BOQ & Technical specification of all assets and specification of work mentioned in Supply order as applicable :-

10. **Schedule A part-1/BOQ**

Ser No	Description of Items	A/U	Qty	Lowest rate quoted by the bidder for the Assets (L1 rate)	Remarks
14.1	Provn of 01 X Cook House And Dining Hall in the AoR of 56 Inf Div	Nos	01		Rate as per BOQ after financial bid opening.
Total amount including GST Rs.					
(Rupees _____ only)					

SPECIAL NOTE : The drawings attached with the bid documents is tentative in nature for quoting of rates only. Refers to size/dimensions of building / structure members can be taken. The contractor is required to provide the following:-

11. Structural design and structural adequacy certificate with design proof checked from Practicing Structural Engineer (Registered)/NIT/Government Engineer College should be submit before commencement of Job.

12. The contractor has to submit a structural suitability certificate as affidavit design has been got verified from NIT/Government Engineer College/Registered structural practicing Engineer and design offered for execution is suitable for the structure.

13. Modification in the design if any to be approved from Chief Engineer and to be submitted before execution for approval of competent authority.

14. Certificate of Min 10 years of structural stability and adequacy to be provided by the contractor. Failure in structure in future will be the responsibility of the contractor wholly.

Consignee. CO, 110 Engineer Regiment, c/o 99 APO.

1. The items para 14 Ser No 14.1. shall be supplied at **56 Mtn DIV AOR..**

2. The entire supply should be completed on or before _____ **2026.**

3. The quantity of items mentioned in this supply order may be increased/decreased due to change in scope of work on the direction of the Commanding Officer, 110 Engineer Regiment. The Commanding Officer reserves the right to add or delete any item based on the assessed cost.

4. **Technical/ Particular Specifications.** The technical specifications of the item of works is given as **Appendix 'A' & 'B'** of the General Conditions of the tender document. These particular specifications are to be read in conjunction with the drawings and any missing detail if any to be provided by the tenderer as per standard engineering practices/ MES SSR/ IS Code to make the shelters fit for use.

5. **Confidentiality.** The Vendor will ensure confidentiality of the project and will not disclose any information, explicit or tacit, gained by them or their representative or agents, while interacting with the Officers of 110 Engineer Regiment. The authorized representative of the Vendor will sign the '**Certificate of Confidentiality**' given at **Appendix 'A'** with the Company Seal duly affixed.
6. **Inspection Authority.** The store/work will be subjected to an inspection to ascertain that they confirm to the specification as laid down in the supply order. The costs incurred during inspection, if any (only for testing of material), will be borne by the Vendor / Supplier.
7. **Raw Materials.** All raw materials used should be new and as per particular specification given in Technical specifications. Test certificate for all raw materials will be submitted by the seller.
8. **Cancellation.** In case the work is not completed within stipulated period, or if defective shelter is made and the defective work/stores is not replaced/ rectified within 30 days from the date of intimation of such defects, the supply order will be cancelled at the sole discretion of Accepting Officer ie, CO, 110 Engineer Regiment either wholly or to the extent of such defaults. If the defect exceeds Bank Guarantee amount and not rectified after one week from the date of encashment of Bank Guarantee, no stores would be accepted and Supply Order shall be cancelled.
9. **Bidder's offer is liable to be rejected if he fails to upload any of the certificate as asked by the buyer.**
10. **Restricted area condition.** All the work need to be restricted area hence vendor has to follow all security guidelines given by user unit and accepting officer time to time work. A copy of the same will be given to vendor prior to commencement of work. All documents & sketches pertaining to work will be returned on completion of work. Only Indian Nationals shall be employed for the work. Vendor is required to obtain necessary security passes from the Army formations for labours and technicians employed for the work.
11. In case of inferior quality store or fake item, Accepting Officer has full right to cancel the Supply Order on quality grounds and due to breach of trust and no representation on this account will be entertained. The vendor will be black-listed for the for six months. All vendors to check the given specifications as per the TS before dispatch on account of quality and quantity as per the SO clauses. **Accepting offr will share the list of rejected items to all units in AOR of 3 Corps.**
12. **This RFP is being issued with no financial commitment and the Buyer reserves the right to cancel/ change or vary any part thereof at any stage. Buyer also reserves the right to withdraw the RFP or Supply Order before delivery of stores, should it become necessary at any stage.**
13. In case of failure of supply of items by due date, the Officer Commanding reserves the right to cancel this Supply Order.

Sd/- x xx xxx
(Rishi Katiyar)
Col
Commanding Officer

COSTED SCHEDULE OF WORKS

Name of Work : Provn of 01 x Cook House and Dining Hall (CHDH) in 56 Inf Div AoR

Job No : OT/EC/3/56/2026-27/2193

S No	Description of Work	Brief Specifications	A/U	Qty	Rate	Amount	Remarks
1	Provn of qty 01 x Cook House Dining Hall (Shelter parts only)	Provn of qty 01 X CHDH (OR), Size 18.0 x 6.10m with 1.50m wide covered verandah, four partition walls to form two internal store rooms, size doors, seven Upvc sliding windows (4'x4'), two turbo ventilators, false ceiling made of 40mm PIR panels, wall cladding made of 50mm PIR panels, roof cladding made of 0.5mm PPGI colour coated sheet and four steel trusses made of portal frame structure. Main column with web-300"200 x 5mm thick and flanges of 128 x 8mm thick, main rafter with web-250"100x 5mm thick and flanges of 100 x 8mm thick and cantilever verandah rafter of same dimension. The four main portal frames are provided at a distance 6.0 mtr centre to centre and interconnected by 'Z/C' purlins at spacing all as given in the drawing No CHDH-OR-01, seven sheets attached. In addition, rear covered verandah of 18.0m x 1.8m with cantilever rafter of web-250"150 x 5mm thick and flanges of 100 x 8mm thick will be provided. Technical Specifications as per Appx 'A' .	Nos	1.00	1075000.00	1075000.00	
2	Provn of 01 x Cook House Dining Hall (Constr mtrl including E/M Items)	Constr mtrl for Cook House and Dining Hall (CHDH). Detailed specification as per Appx 'B' . List of elect items and constr mtrl is as per Annx - I	Nos	1.00	1379109.00	1379109.00	
3	Provn of 01 x Cook House Dining Hall (Furniture Part)	3.1 Provn of Chair Writing as per tech specifications given in Appx 'C' . 3.2 Provn of Writing Table as per tech specifications given in Appx 'C' . 3.3 Provn of Dining Chair as per tech specifications given in Appx 'C' . 3.4 Provn of Dining Table as per tech specifications given in Appx 'C' . 3.5 Provn of Stainless Steel Table as per tech specifications given in Appx 'C' . 3.6 Provn of Water Dispensor as per tech specification given in Appx 'C' . 3.7 Provn of Powder based fire extinguisher as per tech specification given in Appx 'C' . 3.8 Provn of Store Rack as per tech specifications given in Appx 'D' . 3.9 Provn of Almirah Steel Large as per tech specification given in Appx E . 3.10 Provn of Hand Held Trolley as per tech specifications given in Appx 'E'	Nos	1.00	4500.00	4500.00	
			Nos	1.00	5500.00	5500.00	
			Nos	36.00	3500.00	126000.00	
			Nos	6.00	7500.00	45000.00	
			Nos	2.00	20000.00	40000.00	
			Nos	1.00	12500.00	12500.00	
			Nos	3.00	3000.00	9000.00	
			Nos	1.00	10000.00	10000.00	
			Nos	1.00	18000.00	18000.00	
			Nos	1.00	10000.00	10000.00	
					Total Rs.	2734609.00	
					Say Rs.	27.34609	Lakh

Appx 'A'

(Ref to Ser No 1 of CS to TS No 1 of Job
No OT/EC/3/56/2026-27/2193)

TECHNICAL SPECIFICATIONS OF COOK HOUSE DINING HALL : SHELTER PART

1. Structure.

1.1 **Structural Frame Work.** The shelter main frame will be of Steel sections made using high tensile hot rolled steel plates having minimum YS of 345 Mpa and conforming to ASTM 572M, Gr 50 or IS 2062:350/Fe 490 from approved primary steel manufacturers such as **SAIL/TATA/JSW/JSPL/ESSAR/Bhushan Steel/Bhushan power & Steel and Uttam Value Steels.** The steel structure will consist Portal frame, cold formed galvanized 'Z'/C' Purlins and Girts, sag rods, bracing rods. All the members will be suitably designed for relevant IS 800:1984 and loading shall be considered as per IS 875 and for earthquake IS : 1893 to be followed. Roof slope will be as per drawings attached.

1.1.1 All Hot Rolled sections, bracing, sag rod should conform to IS : 2062 : Fe 410. Minimum metal thickness for Hot Rolled will be as per drawing attached. Such steel will be procured from approved manufacturers.

1.1.2 Secondary members comprised of mainly purlins and girts which are cold formed for 'Z' and 'C' shapes and will conform to ASTM A653M, Class I with minimum YS of 345 Mpa having zinc coating mass of 275 GSM. Elongated (oval shaped) holes to be provided for fixing with cleats to cater for some allowances/ adjustments at the site.

1.1.3 Finishes.

1.1.3.1 Surface preparation shall comprise of abrasive blast cleaning to minimum SA-2.5 SIS-055900 near - white blast cleaning on both external and internal surfaces.

1.1.3.2 Structural steel shall be provided with two coats of polyamide cured epoxy zinc phosphate grey primer of DFT 35 microns per coat of approved make conforming to IS 2074-1992 as primer.

1.1.3.3 Finishing two coats shall be provided with acrylic polyurethane cured - aliphatic isocyanate having DFT of 32 to 35 micron per coat and shall be procured from approved manufacturers. Finish coat to be applied by spray or brush in approved colour at factory itself and final finishing paint to be done at the place of delivery to cater for any scratches. **and final finishing paint to be done at the place of delivery to cater for any scratches or provide approved paint - 10 ltr with 7" foam roller brush - 02 Nos and 3" painting brush - 02 Nos.**

1.1.3.4 **Side Runners.** Side runners connecting the portal frame to be provided only at the top of the eaves and not at the bottom to avoid hassles while fitting the door frames. To be of same colour as of the outer colour of wall panels. The connecting plates of these side runners to be designed in such a way that there is no gap between the column flange and inner surface of the side runner to ensure proper alignment with wall panels.

1.1.3.5 **Cleats.** Cleats provided on the rafters and columns for fixing 'Z'/C' purlins should be made of MS plates, 8mm thick minimum along with suitable nuts and bolts only. **No** self tightening screws to be used for fitting these cleats.

1.2 **Fabrication.** All Fabrication work shall be done in accordance with IS: 800 and IS:7215 read in conjunction with relevant codes mentioned therein.

1.2.1 Mild steel rolled sections and plates shall be cut by sheering/plazma cutting and grinding the surfaces to true sizes and shapes. Gas cutting of mild steel may be permitted by the Consignee, provided that every cut face and edge is smoothed by grinding operation.

1.2.2 Extensive use of templates shall be made in doing fabrication work. Templates shall be clean and should have true surfaces prepared for every successive use.

1.2.3 All material shall be straight and free from twist and bends unless required to be curvilinear in form. If necessary the material shall be straightened and/or flattened/ by pressure.

1.3 **Holing.** All holes shall be made at right angles to the surfaces of the member. Holes shall be clean cut without any torn or jagged edges. Holes shall be done by drilling. Punching shall not be resorted to, unless previously approved by the Consignee. In any case, punching of holes in materials having a thickness in excess of the connector diameter, or, for materials thicker than 16 mm, the hole shall be punched 3 mm less in diameter than the required size and then reamed to the full size. Holes shall not be formed or enlarged by burning or gas cutting under any circumstances.

1.4 **Welding.** In general only Automatic submerged arc welding shall be used for fabrication. Subject to approval of the consignee, metal inert gas welding can be used for short length where access to the location of the weld does not permit submerged arc welding. For end plates MIG welding can be used. The welding work will conform to IS: 816/AWSD1.1, unless otherwise specified.

1.4.1 All welds shall be made only by welders and welding operators who have been properly trained and previously qualified by tests to perform the type of work required as prescribed in the relevant applicable standards.

1.4.2 All welds shall be free from defects like blow holes, slag inclusions, lack of penetrations, undercutting, cracks, etc. all welds shall be cleaned of slag or flux and show uniform sections, smoothness of weld metal, feather edges without overlap and freedom from porosity.

1.4.3 Members to be joined by fillet welding shall be brought and held as close together as possible and in no event shall be separated by more than 3 mm. If the separation is 1.2 mm or greater, the fillet weld size shall be increased by the amount of separation. This shall only apply in the case of continuous welds. The fit-up of joints at contact surfaces which are not completely sealed by welds shall be close enough to exclude water after painting.

1.4.4 The separation between fraying surfaces of lap joints and butt joints with backing plate shall not exceed 1.5 mm. Abutting parts to be butt welded shall be carefully aligned and the correct root gap maintained throughout the welding operation. Misalignments greater than 25 percent of the thinner plate or 3 mm whichever is smaller shall be corrected and in making the correction the parts shall not be drawn into a slope sharper than 2 degrees.

1.4.5 No welding shall be done when the surface of the members is wet, nor during periods of high winds unless the welding operator and the work are properly protected. In joints connected by fillet welds, the minimum sizes of single run fillet welds or first runs and minimum full sizes of fillet welds shall conform to the requirements of IS:823. Fillet welds larger than 8mm shall be made with two or more passes.

1.5 **Inspection of Welds.**

1.5.1 All welds shall be inspected for flaws by any of the methods described in these Specifications, and as per IS: 822.

1.5.2 The Suppliers shall arrange for all tests as called for in the schedule of quantities, at his own cost.

1.5.3 In case the tests uncover defective work, such tests shall be at the Suppliers cost and the Supplier shall correct such defects at his own cost, and prove the soundness of rectified work.

1.5.4 The correction of defective welds shall be carried out as directed by the Consignee without damaging the parent metal. When a crack in the weld is removed, magnetic particle inspection or any other equally positive means as prescribed by the Engineer shall be used to ensure that the whole of the crack and material up to 25 mm beyond each end of the crack has been removed.

1.6 **Material Testing.** If mill test reports are not available for any steel materials, the same shall be got tested by the Supplier to the satisfaction of the Consignee to demonstrate conformity with the relevant specification.

1.7 **Tests on Welds.**

1.7.1 **Ultrasonic Test.** Ultrasonic test shall be conducted by the supplier to detect gas inclusion (pores), slag inclusion, shallow welds, cracks, lamination and friability etc for the fillet joints.

1.7.2 **Dye Penetration Test.** Where welds are required to be examined by dye penetration inspection method, such tests shall be carried out in accordance with IS: 3658.

1.7.3 **Radiographic Inspection** Whether instructed by the Consignee, or not, all 'Butt' welds shall be fully inspected by radiographic examination method. Such examination shall be done in accordance with the INSIS; recommendations of IS: 1182.

2. **PIR Insulated Wall Panels.**

2.1 All materials required for the manufacture of shelter shall be new and shall comply with relevant Bureau of Indian Standard specification and bear the Aprk.

2.2 The main walls shall be made up of **50 mm thick** and partition walls shall be made up of 40mm thick Hybrid (Poly-isocyanate) PIR panel sandwiched between PPGI sheets. 0.50 mm thick hot dipped galvanized steel sheets (Total coated thickness 0.54mm) on both side with 50/40 mm thick layer of foam (Density 40 Kg $\pm$ 2 Kg /Cum) insulation sandwiched panels. The wall panels shall be provided with cam lock systems for inter locking. The PIR panel sandwiched between PPGI sheets shall have fire retarding and self-extinguishing properties as per international standard B2 DIN 410 3 Part I.

3. **Wall Panels (Internal/External)** All walls shall be made of PIR insulated panel with insulation core polyisocyanurate foam (PIR) insulation. **PIR insulated panel shall be 50 mm thick with min density 40 Kg/Cum or more.** The panel insulation material in the panel shall have fire retarding and self-extinguishing properties as per international standard. There should not be any gap between wall panels and wall and roof by provision of suitable joining arrangement. Other details of wall panels are as follows.

3.1 All material required for the manufacture of shelter shall be new and shall comply with relevant Bureau of Indian Standard Specification.

3.2 The bulk density of PIR panel insulation should be min **40 Kg /Cum or more insulation core (PIR) polyisocyanurate foam** and the dispensing machinery should be kept with a PLC controlled panel for monitoring and controlling the injection rate to assure specified uniform density requirements. The total thickness of the finished composite panel should be 51 mm. The tolerance in the panel can only be on the plus side

3.3 The outer and inner colour coated (RAL-1001) HI skin made from hot dipped galvanized steel of the panels should be 0.50 mm thick TCT (Total coated thickness). Each panel should be supplied pre-painted with a colour coating of 50 microns of

architectural polyester on a minimum 120 GSM base of Zinc coating on the finished surface only for protection against scratches during handling and transportation. Base metal of GI skin CRCA should be as per **IS-513-2016 (reaffirmed 2021)** and grade of zinc coating will be 120 GSM as per **IS-277-2018 (reaffirmed 2022)**

3.4 The pre-coated GI sheet skin should have minimum coating of 4-5 micron epoxy primer and 25 micron polyester top coat on the finished surface and 7-8 micron primer Alkyd back coat which is bounded to the (PIR) polyisocyanurate foam. The pre coated GI sheet should conform to **IS-14246-2013 (reaffirmed 2018)** with manufacture test certificates conforming to specification.

3.5 The PIR insulated core of these composite panels shall have the following properties :-

- 3.5.1 Density (IS 11239: Part 2) - min **40 Kg/ m³ or more**
- 3.5.2 Panel wt should not be exceed - 10.99 Kg/ m³
- 3.5.3 U value of PIR panel will not exceed – 0.30 w/m²K
- 3.5.4 Flammability - BS2D0 (min)
- 3.5.5 Blowing Agent – Non CFC & Non HCFC
- 3.5.6 Compressive Strength at 10 % deformation - 115 KN/m²
- 3.5.7 Tensile Strength (Adhesion Strength) (IS 16203) – 2.5 Kg/Sqcm
- 3.5.8 K Value (IS 16203) - Equal or less than 0.023W/mk at 10° temp & 0.025 w/mk at 25° temp
- 3.5.9 Water absorption (ISO 2986) – 8-5 ng/Pasm(max)
- 3.5.10 Close Cell Content (IS 11239 Part 5) – Min 85 %
- 3.5.11 Air and water permeability – No absorption of vapor or water
- 3.5.12 Horizontal burning characteristics – max 25 mm
- 3.5.13 Dimensional Stability at 100° C and - 40° C Overall max 2.0 % (IS 11239:Part 3)
- 3.5.14 Core Type – Self blended hybrid insulation core
- 3.5.15 Composition – Foam based closed cell structure
- 3.5.16 Green Mtrls – 100 % recyclable
- 3.5.17 Fire certification – Fire safe test report showing integration, insulation as per EN13501-2
- 3.5.18 Effect of Fire – No loss of insulation core to be ensured
- 3.5.19 **Smoke generation**
 - 3.5.19.1 Smoke generation rate at 12 min <0.07g/s
 - 3.5.19.2 Max smoke generation rate <0.23 g/s
 - 3.5.19.3 Total smoke generated < 60 g
- 3.5.20 Joints in Panel – Joint less (Length upto 11.8 Mtr)
- 3.5.21 **Ash Content**
 - 3.5.21.1 The core material should have minimum ash content of 90 % when tested without adhesive
 - 3.5.21.2 The core material has max heat combustion of 2.0 kj/g (860 BTU/lb) when tested without adhesive

3.5.21.3 The core material should not show visible flaming when tested at an applied heat flux of 50 kW /m² in air enriched to 40 % oxygen without adhesive or facers.

3.5.22 **Note :-** All panels will be manufactured in single piece as per approved layout drawings using the above materials and manufacturing process. Continuous line Polyisocyanurate foam Panels, being used in wall /ceiling /any other location will be tested for its physical /chemical /mechanical properties. Test Certificate for PIR insulation given by CSIR -CBRI Roorkee . Certificate will also give out conformity to the following IS – 12436 -1988 (reaff 2022)IS 11239 part 2,4,5 11 & 12 1985- 1988 (reaff 2019) and flammability test certification conforming EN13501-1 or 476 Part 20 to 22 from CSIR-CBRI Roorkee /IAS labs/NABL will be produced by supplier to consignee at the time of supply of each asset. (Make –Modern Prefab/Llyod/Mahindra Comp/Tata Adv Comp)

3.6 2 mm thick MS Sheet 'C' channel of size 55 mm x 40 mm at the bottom (Floor level) and Top shall be provided to side and fit the wall panels. Suitable arrangement will be made to fix the 'C' channel to floor at bottom and top to structural members.

3.7 All panels should be moulded in place using the above in-situ process through continuous line manufacturing process after placing them in a hydraulic process. A spare panel will be supplied along with bulk supply of panels for testing the designed parameters of the incorporated panels in the shelters.

3.8 The purchaser can get quality testing of any panel from the lot supplied to ensure quality control as per given specifications. Cost of quality testing will be borne by supplier.

4. **Roof / Wall PPGI Sheets.** Roof sheet will be of PPGI 0.50 mm with guard film. The PPGI sheet shall have minimum coating of 4-5 micron epoxy primer and 18-20 micron polyester top coat. The top of 0.50 mm PPGI will have a guard film of 30-40 micron on the finished surface only for protection against scratches. The Pre coated GI sheet will confirm to IS-14246-1995. Base metal of PPGI will be of CRCA as per IS-513 and galvanised as per IS-277 having zinc coating mass of 120 GSM. Same type of sheets will be utilized for walling in case of store shelter. **Minimum 0.30m all around projection and 0.15m overlapping to be provide and same should be considered in roof sheet quantity. The length of purlins also extend upto projection area.**

5. **False Ceiling.** The false ceiling will be provided in the PIR shelters (including back side verandah and excluding front verandah) as per details given below :-

5.1 **False Ceiling.** False ceiling with white colour matching with inner panels shall be provided to the entire shelter including veranda. The false ceiling shall be made of **40mm thick composite PIR panel** sandwiched between PPGI sheets, 0.5 mm thick sheet on inner & outer skin. False ceiling shall be fixed on L & T system. The Panels will be made of 0.5 mm thick PPGI on both side with 30mm thick layer of PIR foam of density of min 40 Kg/ cum or more as Insulation. The false ceiling panels will have tongue & groove arrangement & shall be interlocked using cam lock systems. Aluminium powder coated flashing of size 40 x 40 x 3mm shall be used to fix all around the wall panel and the false ceiling panel for support & joining by means or rivets at minimum distance of 200mm on both the faces.

6. **Doors for PIR Shelters.** Doors will be made out of **PIR panel** of 25 mm thick fixed with aluminum frame single leaf with fixed arrangement. Door frame (Chowkhat) will be made of aluminum section. Door shutter will be fixed with 2 Nos. aluminum AL drop, aluminum tower bolt at the top and at the bottom on each shutter, one sliding bolt and one bow handle with fixing arrangement. Refer attached giving details of door frames for all shelters.

7. **Fly Proofing.** All external doors of OR Living would be provided with suitable fly proof door panels and frame as per details and Drg att as FPD 1/1.

8. **UPVC Windows**

8.1 Providing and fixing Factory-made **uPVC** three track sliding window with white colour frame shall be made from extruded **uPVC** section having overall dimensions

(1200x1200mm) as shown in the **drawing** and having minimum thickness of 2.0 mm (+/- 0.2 mm). All frames & sash shall be reinforced with galvanized steel having minimum thickness of 1.2 mm + 0.2 mm. The frame shall be fabricated in factory as per nomenclature of the item and as directed. **uPVC** track sliding window profile with hollow section with multi chamber, Frame-112 (112mm x 52mm), sliding sash (42mm x 67mm), sliding mesh sash (42mm x 57mm) Sliding Interlock (One vertical length in each shutters) (45.5mm x 28mm), Glazing bead (24.5mm x 16mm) will be mitred cut and fusion welded at all corners, including drilling of holes for fixing hardware and drainage of water etc., making arrangement for fixing of hardware, EPDM gasket, 1.2 +/-0.2mm thick galvanized steel Frame (30mm x 18mm)*2, sliding sash (30mm x 18mm), sliding mesh sash (26mm x 10mm) profile to be inserted in required profile, frame will be fixed to the wall with 8mm x 100mm long fasteners. Fly proof shelters also to be part of window for mosquitos and fly stoppage. **Hardware:** Touch Lock , steel roller system, Stainless steel wire cloth 0.36 mm nominal dia of wire and average width of aperture 1.40 mm and fixed with stainless steel wire staples, 4mm thick tinted glass fixed with gasket, track on bottom rail for three track sliding, wool pine for sliding sash and interlock.

8.2 In case of missing details all needs to be done as good engineering practice by the contractor. Location and sizes of windows and ventilators are given in drawing.

9. **Ventilators (V).** Two way sliding aluminium ventilators of size 600 mm x 600 mm with all fitments and fixed stainless steel wire mesh on outer side to be provided. The ventilator frame and ventilator shutter will be manufactured using suitable Hindalco/ Jindal powder coated (60-70 micron) aluminium extruded sections 6063: T6 Al alloy having minimum thickness of 1.5mm. The ventilator will have double shutter having glazing with 5 mm thick toughened clear glass. The fly proof SS wire mesh of dia 28 guage and aperture size of 1.5mm 304 Grade will be fixed on outside with Aluminium flats. Necessary locking arrangement will be made for the ventilator shutters. Refer drawing attached with each type of shelter giving details of ventilator frames.

10. **Hardware and Fasteners.** All anchor bolts/foundation bolts will confirm to IS: 2062 Grade 4.6. Connection bolts and fasteners will be as per IS:1364 latest Grade 8.8. One washer will be supplied with each bolt confirming to IS : 2016.

11. **Sunshades.** Suitable sunshades made out of PPGI sheets and angle iron frame will be provided to all external windows and doors not covered in verandah. The min projection for the sunshades will be 450 mm and 300 mm wider than the width of the opening. Necessary arrangement shall be made to fix angle iron frame on to the iron members. Frame shall not be fixed on to the wall panels.

12. **Wind Tie.** Flat iron of size 30mm x 3mm thick covering the entire length of the shelter including roof sheet projections will be provided **06 Nos per CHDH**. Elongated holes on the strips to enable fixing of 'J' hooks with two additional bitumen washers at a spacing in alignment with ridge of PPGI sheets.

13. **Sky lighting in Kitchen Area.** For maximum utilization of natural light, four Nos transparent acrylic sheet matching with the pattern of PPGI roof sheet shall be provided on the kitchen roof.

14. **Enclosure of Rear Verandah.** One bay of rear verandah behind the kitchen area to be enclosed with same material as of walls. Two windows shall be provided on the rear side and double leaf door shutter as in drawing for entry.

15. **Turbo Ventilators for CHDH.** 02 Nos Turbo Ventilator for each CHDH should be supplied with necessary fitments/ interface and other accessories to be fitted on the roof in place of roof cladding. Technical specification of Turbo Ventilator (SS 304) is as under:-

MOC of Vanes	SS 304 or superior
MOC of Top Plate	SS 304 or superior
MOC Bottom of Ring	SS 304 or superior
Weight of Ventilator	12 Kg. approximately +/-10

Height of Ventilator	410mm ± 5 mm
Center Width of Ventilator	735mm ± 5 mm
Dia of Top Plate	500mm ± 5 mm
Dia of Bottom Ring	600mm ± 15 mm
Nos of Vanes	42 Nos Rolled Formed
Thickness of Vanes	0.45 mm
Nos of Bearing	02 Nos
Bearing Type	6001 ZZ Permanently Lubricated & Sealed
Rivets	SS 304

16. **Electrification.** Electrical points for lights/fans/power shall be provided as given in the store list attached in **Annexure-I**. In addition the main switchboard/MCB DBs comprising of SP/ DP MCB's shall be provided. All electrical items (MCB/Switches/Cables etc) shall be ISI marked and conforming to relevant IS specifications.

17. **Fan Hooks.** Hooks for fixing of ceiling fans are to be provided with supporting rod connected with the 'T' joint supporting the false ceiling panels with proper arrangements for securing it with help of nuts and bolts along with safety locking pin.

18. **Exhaust Fans.** Where ever exhaust fans are provided, a proper cut/opening to be provided for accurate fitting of exhaust fans and to be totally sealed with necessary sealant with nice finish to avoid exposure of PUF between the outer and inner PPGI sheets of the panels.

19. **Base Plates.** Necessary base plates be provided as per the number of columns of the shelter for proper alignment of the anchor bolts.

20. **Erection tools.** One set for each shelter. Sample set of erection tools will be produced to inspection team and approved during pilot sample inspection. Each set of erection tools will have items as per details given below ISI marked :-

- | | | |
|--------|--|----------|
| 20.1. | Electric Hand Drill Machine 10mm (Make : Bosch/Dewalt/Hira/Hammer) | - 01 No |
| 20.2. | Drill bits 9/64" in packet of 10 Nos (Make : Bosch/Harisson/Power Tech) | - 01 Pkt |
| 20.3. | Drill bits 1/8" in packet of 10 Nos (Make : Bosch/Harisson/Power Tech) | - 01 Pkt |
| 20.4. | Drill bits 10 mm in pkt. (Make : Bosch/Craftsman/Dewalt) | - 01 Pkt |
| 20.5. | Screw driver with insulated plastic handle 12" (Make : Taparia/Estmon/Pie) | - 01 No |
| 20.6. | Screw driver with insulated plastic handle 15" (Make : Taparia/Estmon/Pie) | - 01 No |
| 20.7. | Ball pin hammer 1 Kg with wooden handle (Make : Taparia/Estmon/Pie) | - 01 No |
| 20.8. | Insulated electric pliers 6". (Make : Taparia/Stanley/Hira) | - 01 No |
| 20.9. | Tile cutter 1400 Watt with 6 spare blades (Bosch/Harisson/Power Tech) | - 01 No |
| 20.10. | Gypsum star screw black in colour of 1" (500 Nos) | - 01 Pkt |
| 20.11. | Gypsum star screw black in colour of 2" (500 Nos) | - 01 Pkt |

21. **User Hand Book.** It will be supplied at the scale of One per shelter. The user hand book will be complete with drawings, all details of all parts of the shelters, fittings and fixtures and instructions for erection with detailed drawings. **The users hand book will be supplied well in advance and will be verified from this department that the parts are sequentially numbered for ease of identification and erection of the shelter mentioned in the hand book. Panels and structural members to have the same marking on them for their easy identification.** Their specifications and quantity are strictly and correctly as per technical specifications and the supplier has not changed the specifications. Based on the checked hand book the store will be supplied and received.

22. **Minor Details.** Details not mentioned in the specification like trim & flashings etc., but required, as per good and sound engineering practice should be provided by the supplier. Necessary modifications with the frame may be carried out to suit the same. Only superior modification will be acceptable, as per standard engineering practice.

23. **Materials Make :-** The quality and sizes of the various types of items provided or used while supply or fabricating shall be conforming to relevant technical specifications. **The make that are**

offered will be intimated for approval of Accepting Officer within 07 days from date of placing of supply order. Once make approved as offered and accepted will be final and all materials supplied under supply order will conform to the approved make offered. In case of any disputes / missing of details on account of specifications / makes, bidders are requested to intimate to tender inviting officer before bid quoting. Once tender is accepted no deviation shall be entertained. In this case decision of Accepting Officer shall be final and binding.

24. Various members and their quantities to be provided with each shelter as given drawings enclosed is tentative and an indicative only. The exact quantities to be provided with each shelter shall be as per actual requirement.

25. **References of drawings.** Following references would be adhered to while selecting the drawings for each type of shelter.

25.1 CHDH(OR) - CHDH OR-01 (SH 1 of 7, 2 of 7, 3 of 7, 4 of 7, 5 of 7, 6 of 7 & 7 of 7)

25.2 Fly Proof Door - FD 1/1

NOTE :-

1. The supplier will be whole and sole responsible to supply and construction of the items required for the asset as per specifications and good engineering practice including minor extras.

2. Firm/Vendor has to provide one technical representative for supervision and rectification of any fabrication issues in structure till erection of complete Shelter.

TECHNICAL SPECIFICATIONS FOR COOK HOUSE DINING HALL

1. **PCC Foundation.** PCC Foundation will be made by using 400 x 200 x 200mm pre cast cement concrete block, in cement mortar 1:6 over 100 mm thick PCC 1:4:8 over 150 mm thick Hard Core all as shown in Drg No 01 of 02 (Fdn details) att. 15mm thick Cement plaster in cement mortar 1:4 will be done on block masonry above ground level on the external surfaces.
2. **RCC Foundation.** 08 Nos x RCC column of size 300 x 450mm with footing of size 1200 x 900mm over 100 mm thick PCC 1 : 4 : 8 over 150 mm thick hard core all as shown in Drg No 01 of 01, (RCC fdn details) att. 16mm dia foundation bolts of main portal frame base plate has to be suitably aligned and placed in the RCC column.
3. **Plinth Protection.** All around the shelter, plinth protection of **750 mm** wide, 75mm thick with PCC 1:3:6 type C1 shall be laid over 100 mm thick hard core over compacted soil.
4. **DPC.** 50mm thick PCC 1: 2: 4: Type B-1 to be provided over cement concrete block masonry wall surface.
5. **Tiled Flooring.** Non skid colour tiles 600x600x10 mm thick laid over 20mm thick screed bed in CM 1:4 over 75mm thick PCC 1:3:6 using 20mm graded stone aggregate over 75mm thick hard core laid over compacted soil.
6. **Pathway.** Chequered Tile of size 250 mm x 250 mm x 25 mm) thick laid over 20mm thick screed bed in CM 1:4 over 75 mm thick PCC 1:4:8 over 100 mm thick hardcore laid over compacted soil.
7. **Steps.** A step shall be constructed out of **PCC solid block** masonry in CM 1:6 in front of the verandah in alignment with each door with riser as 200mm and treads as 300mm.
8. **Service counter.** Service counter will be made by using 400mm x 200mm x 200mm PCC solid blocks in CM 1:6 over 100mm thick PCC 1:4:8 cement plaster in CM 1:4 will be done on block masonry above ground level on both the sides. Top surface of the service counter shall be provided with granite stone slab as per size required.
9. **Water Supply Fittings.** One HDPE storage tank of capacity 1000 ltrs with staging and connections will be provided with details as given under :-
 - 9.1. Staging for water storage tank of size 1.50m x 1.50m x 4.50m height, shall be made of MS Al 75x75x5mm vertical posts (04 Nos), 50x50x5mm diagonal members and 75x75x5mm horizontal members. MS chequered top plate made of 4.00 mm thick, surrounding & diagonally supported from bottom with MS Al 75x75x5mm thick. MS Flat Iron 40x5mm thick will be provided at the center joint of the MS plate. At the bottom of each vertical post 100mm x100mm x 6mm thick MS plate shall be provided for stable of the staging complete as per drawing.
 - 9.2. PPR pipe 15/20 mm diameter with fitting as per store list.
10. **Internal Water Supply Fittings.**
 - 10.1. **Pipe and fitting.** All pipes & fitting for internal water supply shall of PPR Pipe. Main outlet pipe from PVC tank not less than 25mm diameter & distribution pipe 15mm diameter shall be provided.
 - 10.2. **CP Brass Pillar Tap.** 15mm diameter chromium plated brass pillar tap shall be provided conforming to IS – 8934 of 1978.

10.3. **CP Brass Bib Tap.** 15mm dia chromium plated brass screwed down bib tap shall be provided conforming to IS – 8931 of 1978.

10.4. **CP Brass Stop Valve.** CP screwed down stop valve shall be provided conforming to IS-781 of 1984.

11. **Manhole.** Manhole shall be made of PCC solid block of size 400mm x 200mm x 200 mm in CM 1: 6. The internal size of manhole shall be of 600x600x600 mm depth in length, breadth and depth respectively (depth vary as per ground condition). The external joints of the hollow block shall be finished flush, and joints of pipe and the masonry shall be made perfectly leak proof. The Man hole shall be built over 100mm thick over PCC 1:4:8 type D2 (40mm graded crushed stone aggregate) over 75mm thick hardcore of broken stone boulders of gauge 63 mm. The channels and benching shall be done in PCC 1: 2: 4 type B1 (20mm graded crushed stone aggregate). The wall of manhole shall be plastered with 15mm thick in CM 1:4 by mixing water proofing compound @ 3% by weight of cement, plastering should be even and smooth without using extra cement. RCC manhole cover shall be provided.

12. **Disposal of waste water.** All outlets from manhole will be connected to a soakage pit. The internal size of soakage pit is 1.20 x 1.20 x 1.20m which is built by PCC solid blocks over 100 mm thick PCC 1:4:8 type D2 by over 75 mm thick hardcore of broken stone boulders of gauge 63 mm. RCC cover shall be provided on soakage pit. The vent pipe with cowl should be provide.

13. **Washing Trough.** Washing trough of size 2.0m length with 0.6m wide shall be constructed with following specification:-

13.1. 3 Nos of 200mm thick PCC block masonry wall pillar of size 0.6m length and 0.8m height shall be constructed over 100mm thick PCC 1:4:8 as fdn concrete over 150 mm hard core .

13.2. RCC Slab trough of size 2.0m x 0.6m x100 mm thk to be constructed over PCC block pillar by using RCC 1:1¹/₂:3 with 8mm dia TMT bars 150mm c/c on both way.

13.3. Glazed/ceramic coloured tiles to be provided completely with tiles of size 300x600x8-9mm thick over 15mm thick rendering in CM 1:4.

13.4. 5 Nos of Bib cock to be provided in washing trough including all necessary GI pipe line complete.

14. **Eaves Gutters and Rain Water Pipes.** PVC sheet eaves gutters , half round 150mm fixed with flat iron frame 25x3mm thick clamp, cleats and 110 mm bore PVC pipes single socketed, in any length with rubber ring joints, fixed in wall with cladding for rain water down pipe complete all as specified and directed.

15. **Earthing.** Earthing complete with galvanized earth plate electrode 600x600x6mm thick, buried directly in ground earth pit not less than 2.25 meters below ground level connected to galvanized earth strip of size 32x6mm by means of bolts, nuts, GI washers GI pipe 40mm dia light grade 7.5m long connected to earth test point including test point complete including earth work excavation in any type of soil PCC (1:3:6) type C1 bed and chamber RCC (1:2:4) type B1 cover 40mm thick including 6mm dia reinforcement and funnel watering pipe etc all as specified.

16. **Electrification.** Electrical points for lights/fans/power shall be provided as given in the store list attached in **Annexure-I**. In addition the main switchboard/MCB DBs comprising of SP/ DP MCB's shall be provided. All electrical items (MCB/Switches/Cables etc) shall be ISI marked and conforming to relevant IS specifications.

17. **Materials Make :-** The quality and sizes of the various types of items provided or used while supply or fabricating shall be conforming to relevant technical specifications. **The make that are offered will be intimated for approval of Accepting Officer within 07 days from date of placing of supply order.** Once make approved as offered and accepted will be final and all materials supplied under supply order will conform to the approved make offered. In case of any disputes /

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missing of details on account of specifications / makes, bidders are requested to intimate to tender inviting officer before bid quoting. Once tender is accepted no deviation shall be entertained. In this case decision of Accepting Officer shall be final and binding.

18. List of constr mtrls for CH/DH shelter is as per att **Annex - I**.

Note :-

1. The supplier will be whole and sole responsible to supply of the items required for the asset as per specifications and good engineering practice including minor extras.
2. Plinth wall height below the ground level has to be as per the ground condition and any variation based on actual soil strata be created by initiating supplementary TS.

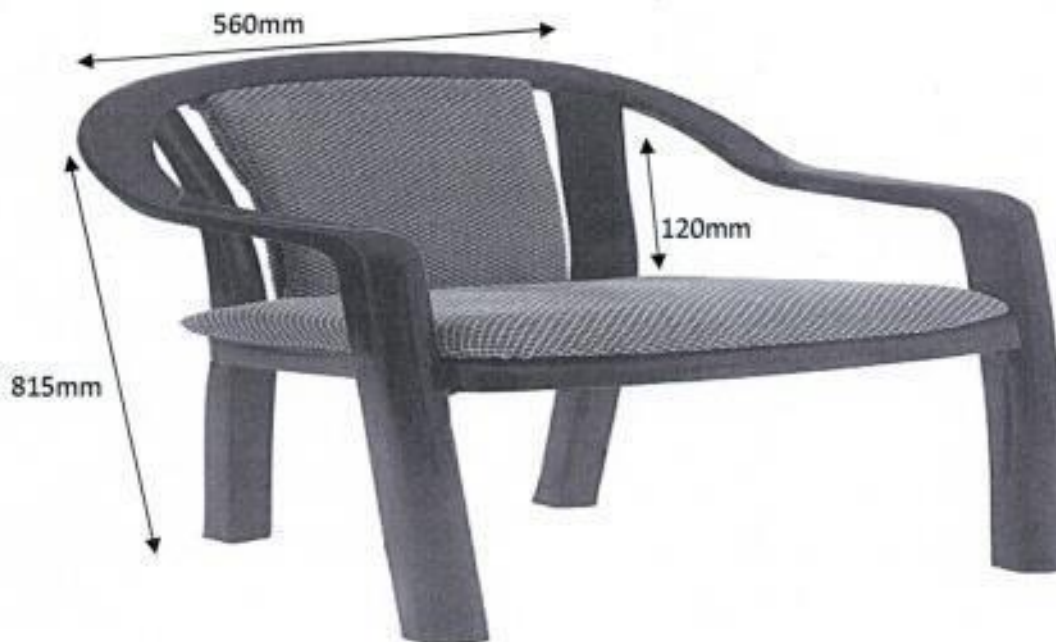
Appx 'C'

(Ref to Ser No 3 of CS to TS No 1 of
Job No OT/EC/3/56/2026-27/2193

TECHNICAL SPECIFICATIONS FOR FURNITURE

1. **Writing Chair.** The Writing Chair will be Lacquer finish, synthetic resin and medium back with arms, seat & back cushion.

- | | | |
|----------------|---|---|
| 1.1. Brand | - | Supreme/Nilkamal/Cello |
| 1.2. Material | - | Plastic |
| 1.3. Color | - | Ornate Rose wood, Brown |
| 1.4. Dimension | - | Height - 815mm, Width – 560mm & Depth – 120mm |
| 1.5. Cushion | - | Back & Seat |



2. **Writing Table :-** Table size of 1145mm x 610mm x 760mm Shall be provided. Table top shall be 19 mm thick prelaminated decorated particle board fixed with hollow section frame with screw@200 mm c/c. The frame work of table shall be of MS hollow section of Size 50X50 mm all as per Drawing. End of leg will be sealed 1.6mm thick with proper welding process to avoid the dust entry. 6mm thick teak wood lipping shall be provided on all exposed edges of MDF duly treated over with French polish as specified. All steel surface will have two coat of synthetic enamel paint over a coat of red oxide. The legs will be fitted with hard PVC Shoe.

3. **Dining Chair :-** Dining chair, Wooden Seat, Pipe Frame with Chrome Plated.



4. **Dining Table :-** Table size of 1800mm x 900mm x 750mm shall be provided. Table top shall be 19 mm thick Pre laminated decorated particle board fixed with hollow section frame with screw@200 mm c/c. The frame work of table shall be MS hollow section of Size 50X50 mm all as per Drawing. End of leg will be sealed 1.6 mm thick with proper welding process to avoid the dust entry. 6mm thick teak wood lapping shall be provided on all exposed edges of MDF duly treated over a with French polish as specified. All steel surface will have two coat of synthetic enamel paint over a coat of red oxide primer. The legs will be fitted with hard PVC Shoe.

5. **Stainless Steel Table :-** Stainless Steel Table of size 1800 x 750 x 760 mm shall be provided. Table top shall be 14 gauge stainless steel sheet on top fixed with 35mm stainless steel square pipe with screw. The frame work of table shall be made of stainless steel tube of size 35 x 35 x 1.25 mm thick all as per drg. Stainless steel wire mesh shall be fixed at the bottom of the table as per drg. The legs will be fitted with hard rubber shoe. Refer SST 1/1 as att.

6. **Bottled Water Dispenser.** Bottled Water dispenser body made of ABS plastic and tank made of Food Grade stainless steel. Capacity of cooling and hot water shall be of 3 litres and 5 litres respectively with 3 temperature taps like Hot, Plain and cold. Heavy duty bottle piercer with bottle holder shall be provided to hold the water bottle of capacity 20 ltrs. Bottom of the body shall consist of a refrigerator of 14 ltrs capacity. Make: Blue star / Usha.



7. **Powder based Fire Extinguisher (Red) - 1** : Dry powder fire extinguisher of 1 Kg Capacity in confirmation with ISO/CE certified with wall mounting holder. It shall be helium leak tested and with 3 years warranty. Make : Ceasefire / Suprema / Eco fire



NOTE :- The supplier will be whole and sole responsible to supply and construction of the items required for construction of the asset as per specifications and good engineering practice including minor extras.

Appx 'D'

(Refer to Ser No 3 of CS - 01 to TS No 1
of Job No OT/EC/3/56/2026-27/2193)

TECHNICAL SPECIFICATIONS FOR STORE RACK

1. **Rack Platform.** Four platforms, each made of angle iron of size 40mm x 40mm x 5mm, welded to form a rigid frame work to hold 4' x 2' ply wood. Suitable holes (two Nos) to be provided at each corner to fit the platform to four vertical angles. Refer drg No. SR-1/1.
2. **Vertical Columns.** Four Nos vertical legs made of angle iron of size 40mm x 40mm x 5mm with a MS plate of size 80mmx80mmx8mm welded at the bottom. Suitable holes at intervals as shown in drg attached be provided for fitting these vertical columns to four rigid platforms with nut & bolts to make a store rack.
3. **Plywood.** 12mm thick BWR wooden Ply of size 8' x 4' board to be taken and cut into four pieces of size 4' x 2' each for optimal utilization and reduction in wastage. Ply to be fixed on rack platform at four places by using flat head self-tightening screws. Plywood to be ISI marked, water resistant, termite resistant and treated with preservative as per IS-5539-1969 ply rating.
4. **Bracing.** Two Nos MS flat of size 40mm x 5mm to be provided at the rear as shown in the drg to act as bracing for the rack. Suitable holes to be provided for fixing the same with nut & bolts.
5. **Finishes.** All steel surfaces of the store rack shall be painted with two coats of anticorrosive red oxide primer at the time of supply. Before application of paint all the steel surface shall be thoroughly cleaned to make the same free of rust, mill scale, dirt, oil etc either by mechanical or chemical means. Finally, the store rack shall be painted with two coats of olive green paint at factory or on site by the supplier.
6. **Hardware.** Suitable nut & bolts of size 10mm to be provided for fitting the platform to the vertical columns. Additional 10 Nos nuts and bolts be provided with each rack as spare. Exact holes to be made accordingly where necessary to avoid any play at a later stage after fitting the rack in one piece.
7. **Minor Details.** Details not mentioned in the specifications but required as per sound and good engineering practice should be provided by the supplier.
8. **Reference of Drawing.** Drawing No 1/1 of Store Rack.

Note: Rubber Shoe Cover will be provided in all the furniture items.

Appx 'E'

(Refer to Ser No of CS - 02 to TS No 1
of Job No OT/EC/3/56/2026-27/2193)

TECHNICAL SPECIFICATIONS FOR FURNITURE

1. **Almirah Steel Large.** Almirah steel large shall as per drawing of size **1985 x 1300 x 485mm** made up of 18mm thick MS sheet. Four number of shelves shall be provided equally spaced to make five compartments. One chromium plated handle 150mm long with 3 way locking device shall be provided. In addition one Hasp and Staple of size 150mm long shall be provided for alternative locking arrangements. All steel surfaces internally and externally to be painted with 2 coats of synthetic enamel paint over a coat of red oxide primer. Refer Drg No ASL 1/1.
2. **Hand Held Trolley.** Body frame 900mm x 600mm size made up of 50mm x 50mm x 6mm MS angle, top of the frame covered with 2mm thick stainless steel sheet and 25mm dia MS round bar covered with rubber sleeves used for handle. 04 Nos of pneumatic scooter tyre with rim and ball bearing shall be provided.

STORE LIST FOR CONSTR OF EACH COOK HOUSE DINING HALL

S No	Items	A/U	Qty Per Set	Remarks
Constr Material				
1	Cement : Portland Pozzolana Cement: PPC cement, IS-1489 (part-2) for calcined clay based PPC and IS 1489 (Part-1) for fly ash-based PPC), packed in 50 Kgs HDPE bags conforming to IS 11652/2000. (MAKE: Ambuja / ACC / ULTRA TECH/ DALMIA/ RAMCO/ BIRLA CCI)	Bags	263.00	
2	Coarse Sand : Sand shall be clean, sharp, angular hard and durable free from adherent coating and shall not contain clay, mica and other impurities. The maximum quantities of clay and fine silt in sand shall not be more than 5% by mass as per IS-383-1970.	Cum	27.00	
3	20mm Stone Aggregates : Stone aggregate shall be crushed stone 20mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	15.00	
4	40mm Stone Aggregates : Stone aggregate shall be crushed stone 40mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	24.00	
5	Hardcore 63 mm Stone Aggregates : Stone aggregate shall be crushed stone not exceeding 63mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	48.00	
6	PCC Solid block 400x200x200mm (+/-10mm) minimum strength 50Kg/Sqcm. PCC shall conform to IS-2185 (Part I) 2005. Solid Block made out of PCC 1:3:6 having the compressive strength 5 KN/Sqmm.	Nos	2209.00	
7	Tiles Coloured Non-skid of size 600 x 600 x 9-10 mm thick.(Make : Kajaria/Orient/Johnson)	Nos	494.00	
8	Glazed ceramic Coloured tiles Non-skid of size 300 x 600 x 7-8mm thick. (Make : Kajaria/Orient/Johnson)	Nos	79.00	
9	Chequered tile 250x250x25 mm	Nos	404.00	
10	8 mm dia TMT bar as per IS 1786-1979, Make : TATA/SAIL/JINDAL	Kgs	140	
11	12 mm dia TMT bar as per IS 1786-1979, Make : TATA/SAIL/JINDAL	Kgs	230	
12	Draper curtain rod 7' long 20mm dia.	Nos	3	
13	Draper curtain rod 4' long 20mm dia.	Nos	6	
14	Bracket with knob for curtain rod	Pairs	9	
15	Granite stone slab 2 x 0.60 mtr having 20-22 mm thick.	Nos	1	

Sanitary fittings items				
16	Wash Hand Basin with CP waste coupling and PVC waste pipe enclosed with stand along with MS Bracket suitable for fixing of WHB. (Make : Hindware/Parryware/ Cera/Hindustan)	Nos	2	
17	Stainless steel kitchen sink with CP waste coupling and waste pipe along with angle base frame. (Make : Hindware/ Parryware/ Cera/ Hindustan)	Set	1	
18	PVC Connection Pipe 450mm long with both side of neat of size 15mm dia	Nos	2	
19	CP Stop Cock 15mm. (Make : Paryware/Prayag/Cera/Ashirwad)	Nos	3	
20	CP Bib Cock Long Body 15mm dia. (Make : Paryware/Prayag/Cera/Ashirwad)	Nos	2	
21	CP Pillar Cock 15mm. (Make : Paryware/Prayag/Cera/Ashirwad)	Nos	2	
22	PVC waste coupling 32mm dia. (Make : Paryware/Prayag/Sona/Ashirwad)	Nos	5	
23	CP Bib Cock 15mm dia. (Make : Paryware/Prayag/Cera/Ashirwad)	Nos	5	
24	PVC waste pipe 32mm dia. (Make : Prince/Om Shree/Finox/Sanmati)	Nos	5	
25	PVC Looking Mirror cabinet type of size 14 x18 inch , white colour with complete fittings. (Make - Prayag/Nilkamal/Plantex)	Nos	2	
26	Acrylic Bathroom corner Shelves set of 3 unbreakable transparent corner wall shelves (8",10",12") for bathroom with fixing accessories complete	Nos	2	
27	Mirror full length 600mm x 2100mm bevelled edge mirror of selected quality glass, mounted on 6mm thick AC building board or commercial plywood and fixed to wooden plugs with chromium plated brass screws and cup washers.	Nos	2	
28	Steel Door foot Matt /grill/ dirt scraper size 600 x900 mm load capacity 300 to 800kg made up of cast iron anlg iron & MS round bar complete all as specified and directed.	Nos	2	
29	PVC sheet eaves gutters , half round 150mm fixed with flat iron frame 25x3mm thick clamp, cleats and 110 mm bore PVC pipes single socketed, in any length with rubber ring joints, fixed in wall with cladding for rain water down pipe complete all as specified and directed.	Rm	40	
Pipe Fitting Items				
30	Staging for water storage tank of size 1.50m x 1.50m x 4.50m height, shall be made of MS Al 75x75x5mm vertical posts (04 Nos), 50x50x5mm diagonal members and 75x75x5mm horizontal members. MS chequered top plate made of 4.00 mm thick, surrounding & diagonally supported from bottom with MS Al 75x75x5mm thick. MS Flat Iron 40x5mm thick will be provided at the center joint of the MS plate. At the bottom of each vertical post 100mm x100mm x 6mm thick MS plate shall be provided for stable of the staging complete as per drawing.	Nos	1.00	

31	Triple layer vertical cylindrical HDPE Water storage tanks, (Make: Sintex/Polycon/Vectus) ISI marked of 1000Ltr capacity shall be provided. Water storage tank of 1000 litres capacity each along with float valve hinged top lid of same material with locking arrangement with all accessories will be provided as per tech specification. Material shall be confirm as per IS specification IS- 12701 : 1996.	Nos	1.00	
32	Tank Nipple 20mm dia with both side check nut & rubber washer, ISI Marked. (Make : Jindal/Hindustan/Nezone)	Nos	2	
33	PVC Float Valve 20mm with complete set, ISI Marked. (Make : Shruti/VNS/Prayag)	Nos	1	
34	PP-R Gate Valve 25mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	4	
35	PP-R pipe (PN-16/DSR 7.4) 25mm dia (3 layers - outer green UV stabilized, middle black and inner layer microbial white colour) ISI marked (Make : SFMC/Jindal/Vectus/ Prince)	Rm	40.00	
36	PP-R pipe (PN-16/DSR 7.4) 20mm dia (3 layers - outer green UV stabilized, middle black and inner layer microbial white colour) ISI marked (Make : SFMC/Jindal/Vectus/ Prince)	Rm	30.00	
37	PP-R pipe union 25mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
38	PP-R pipe socket 25mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
39	PP-R pipe elbow 25mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
40	PP-R pipe Tee 25mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
41	PP-R Reduciton Tee 25/20 (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
42	PP-R pipe Tee 20mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
43	PP-R pipe elbow 20mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
44	PP-R Reducer elbow 25/20 (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
45	PP-R pipe Tank nipple 20mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	2.00	
46	PP-R pipe socket 20mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	10.00	
47	PP-R wall clamps 20mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	20.00	
48	PP-R wall clamps 25mm (Make : SFMC/Jindal/Vectus/ Prince)	Nos	20.00	
49	PPR end cap 25 mm	Nos	4.00	
50	PPR end cap 20 mm	Nos	10.00	
51	PPR socket male brass threaded 20 mm	Nos	10.00	
52	PPR elbow female brass threaded 20 mm	Nos	10.00	
53	PPR socket female brass threaded 20 mm	Nos	10.00	
54	PPR socket female brass Socket 20 mm -1/2"	Nos	10.00	
51	PP-R Pipe 50 mm Dia Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Rm	40.00	
52	PP-R Pipe 40 mm Dia Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Rm	30.00	

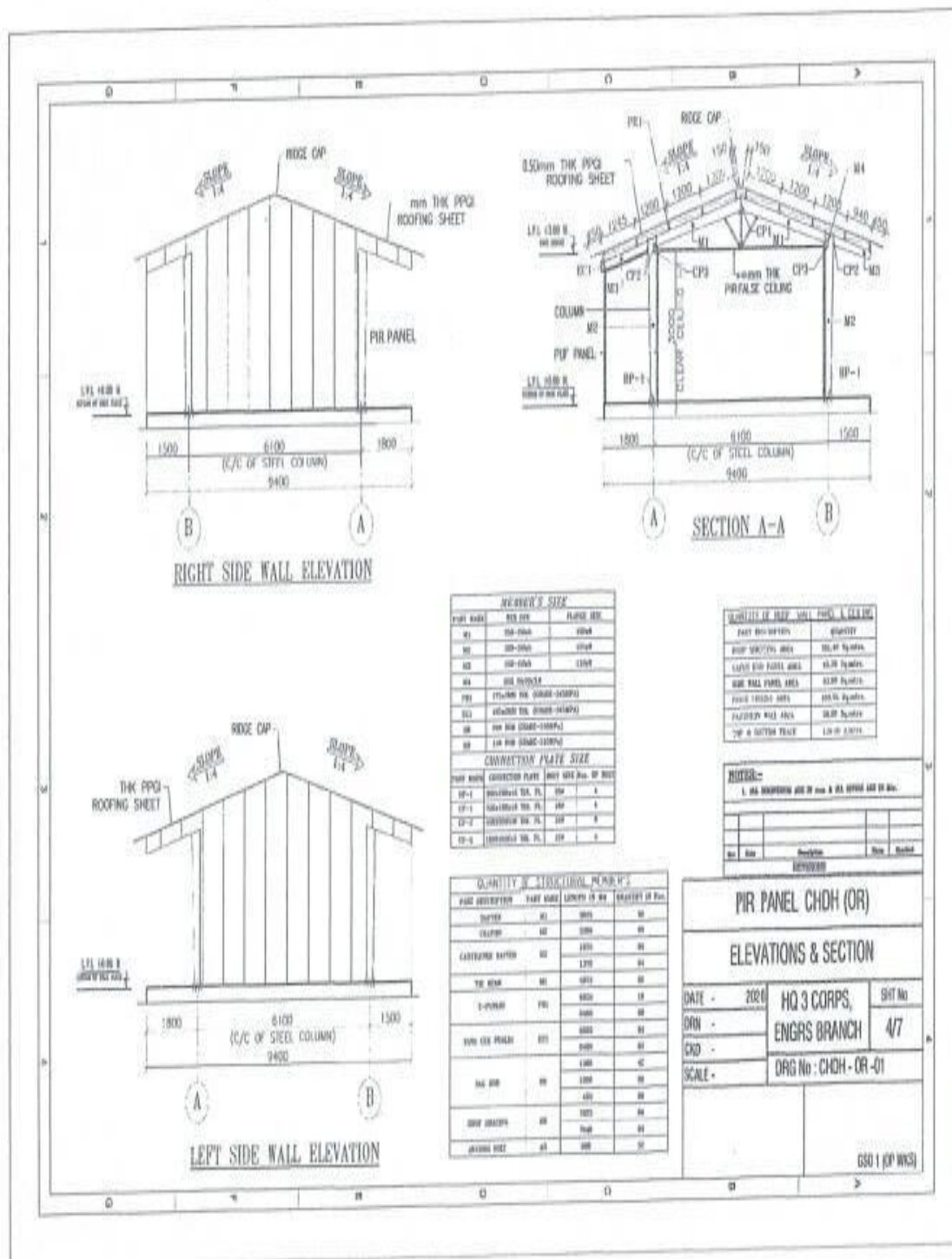
53	PP-R pipe 32mm dia Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	RM	15.00	
54	PP-R Elbow 90° 50 mm Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Nos	3.00	
55	Socket PP-R 50mm Dia Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Nos	12.00	
56	PP-R Reducer 50 mm to 40mm dia Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Nos	2.00	
57	PP-R Elbow 90° 40 mm Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Nos	3.00	
58	Socket PP-R 50mm Dia Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Nos	10.00	
59	PP-R Reducer 40 mm to 32 mm dia Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Nos	2.00	
60	PP-R Elbow 90° 32 mm Make : SFMC/ Jindal/ Welflow/ Reliance/ Vectus/ Prince	Nos	4.00	
61	MCB 63 Amp 230 volt make -Crompton / Bajaj/Philips	Nos	1.00	
62	1 HP Booster pump-01 Nos , Auto starter 1 HP -01 nos ,MCB DP16 -01 Nos ,DP Main Switch -01 Nos ,2.5 Sqmm Copper Wire-40 mtr ,NRV -01 Nos as per ps clause complete all as specified and directed	Nos	1.00	
Sewage Pipe Fitting Items				
63	PVC Pipe of 110mm dia, 6 mtr long with collar for sewage disposal to septic tank, ISI marked. (Make : Prince/Om Shree/Finolex/Sanmati)	Nos	3	
64	PVC Pipe of 75mm dia 6' long with collar for air extension of sewage tank, ISI Marked. (Make : Prince/Om Shree/Finolex/Sanmati)	Nos	2	
65	PVC Cowl 100mm dia for air extension pipe. (Make : Prince/Comet/Finolex/Sanmati)	Nos	1	
66	PVC S - Trap 100mm, dia ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	1	
67	PVC Elbow 100mm dia, ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	2	
68	PVC Tee 100mm dia, ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	2	
69	PVC Coupler 100mm dia, ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	2	
70	PVC Bend 100mm dia, ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	2	
71	PVC Floor trap 75 mm dia with grating, ISI Marked (Make : Prince/Comet/Finolex/Sanmati)	Nos	3	
72	PVC Tee 75mm dia, ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	2	
73	PVC Elbow 75mm dia, ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	1	
74	PVC Coupler 75mm dia, ISI Marked. (Make : Prince/Comet/Finolex/Sanmati)	Nos	2	
75	Water Proofing Compound. (Make : Perma Plast-O-Proof Std)	Kgs	5	
76	White Cement. (Make : Birla Gold)	Kgs	15	
77	Lime unslaked of best quality. (Make : Tata/Lakshmi)	Kgs	15	

Electrical Items				
78	Surface and wall mounted LED tube rod 15 watt, 1600 lumens output with suitable fitting frame. Frame shall be powder coated with aesthetically designed dual colour ends caps with bipin holder, connector block complete. (Make : Bajaj/Havells/Philips)	Nos	11	
79	LED 12 W Rechargeable Emergency Inverter LED Bulb complete with inbuilt Li-ion battery, B22 base, all accessories	Nos	2	
80	Pixel bulk head fittings in deep drawn anodized aluminium body, epoxy white powder coated and heat resistant prismatic glass cover with specially designed gasket with LED lamp 1 x 5 watt. (Make : Bajaj/Havells/Philips)	Nos	3	
81	Ceiling fan 1200mm sweep with electronic regulator conforms to IS : 2312, ISI marked with double ball bearing with electronic regulator MS Fan hooks with MS Clamps bolts and nuts. (Make : Bajaj/Crompton/Havells/Usha)	Nos	6	
82	Pedestal fan 400mm sweep ISI marked with heavy duty speed motor(Make - Usha/Baja/Havells/Orient)	Nos	4	
83	Exhaust fan 300mm sweep, ISI marked with heavy speed motor and louvers with internal side wire mesh . (Make : Khaitan /Usha / Bajaj / Havells /Orient).	Nos	1	
84	PVC indulated, heavy duty, 1100 volt grade cable with aluminium conductor, twin core, 10 sqmm served with sheathing of PVC tape and overall PVC sheathed, flat or circular. (Make : Finolex/Havells/ Polycab/Plaza)	Rm	100	
85	Porcelain insulating hangers	Nos	100	
86	PVC sheathed unarmoured stranded copper conductor cable of size 1.0 sqmm dia conform to IS 694/1900 (green in colour of 90 Mtr Roll). (Make : Finolex/Havells/Polycab/Plaza)	Roll	2	
87	PVC sheathed unarmoured stranded copper conductor cable of size 1.5 sqmm dia conform to IS 694/1990 (Green in colour of 90 Mtr Roll). (Make : Finolex/Havells/Polycab)	Roll	4	
88	PVC sheathed unarmoured stranded copper conductor cable of size 2.5 sqmm dia conform to IS 694/1990 (Green in colour of 90 Mtr Roll) (Make : Finolex/Havells/Polycab)	Roll	2	
89	PVC sheathed unarmoured stranded copper conductor cable of size 4 sqmm dia conform to IS 694/1990 (Green in colour of 90 Mtr Roll) (Make : Finolex/Havells/Polycab)	Roll	2	
90	Modular switch piano flush type 6 Amp, 240 volts single pole one way. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	24	
91	Modular switch piano flush type 16 Amp, 240 volts single pole one way. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	6	
92	Modular socket outlet multipurpose 2/3 pin 6 Amps flush button type. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	3	
93	Modular socket outlet multipurpose 2 in 1, 6 and 16 Amps. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	4	
94	Surface mounting plastic boxes suitable for modular switches 12 module of size 8" x 6". (Make : Havells/Bajaj/Anchor/Legrand)	Nos	4	

95	Surface mounting plastic boxes suitable for modular switches 8 module of size 9" x 3". (Make : Havells/Bajaj/Anchor/Legrand)	Nos	6	
96	Surface mounting plastic boxes suitable for modular switches 6 module of size 8" x 3". (Make : Havells/Bajaj/Anchor/Legrand)	Nos	6	
97	Surface mounting plastic boxes suitable for modular switches 4 module of size 5" x 3". (Make : Havells/Bajaj/Anchor/Legrand)	Nos	12	
98	Combined cover plate with frame (Inner & Outer) white suitable for modular switch/socket of 12 module. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	2	
99	Combined cover plate with frame (Inner & Outer) white suitable for modular switch/socket of 8 module. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	6	
100	Combined cover plate with frame (Inner & Outer) white suitable for modular switch/socket of 6 module. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	6	
101	Combined cover plate with frame (Inner & Outer) white suitable for modular switch/socket of 4 module. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	12	
102	Blanking plate for modular switch/socket. (Make : Havells/Bajaj/Anchor/Legrand)	Nos	20	
103	Ceiling rose, surface bakelite, 65 x 50mm, 2 to 3 terminals. (Make : Anchor/ Crabtree/Havells/Bajaj/Anchor/Legrand)	Nos	25	
104	Flexible pipe PVC heavy duty 20 mm dia	Mtr	10	
105	PVC pipe 20mm dia, heavy duty, ISI marked. (Make : Polycab/Suncab/Indoplast/Surya)	Rm	50	
106	PVC elbow for 20mm dia PVC pipe. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
107	PVC space bar saddles for 20mm dia PVC pipe. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	50	
108	PVC coupler for 20mm dia PVC pipe. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
109	PVC Tee for 20mm dia PVC pipe. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	5	
110	PVC casing capping 25mm wide, not less than 2mm thick. (Make : Polycab/Suncab/Indoplast/Surya)	Rm	50	
111	PVC coupler for 25mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
112	PVC Tee for 25mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
113	PVC sockets for 25mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
114	PVC elbow for 25mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
115	PVC Corner bend for 25mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
116	PVC casing capping 32mm wide, not less than 2mm thick. (Make : Polycab/Suncab/Indoplast/Surya)	Rm	40	
117	PVC coupler for 32mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
118	PVC Tee for 32mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	
119	PVC elbow for 32mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	10	

120	PVC Corner bend for 32mm casing and capping. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	8	
121	PVC junction box two way 20mm dia. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	4	
122	PVC junction box three way 20mm dia. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	8	
123	PVC junction box four way 20mm dia. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	8	
124	PVC junction linker, 75 x 75mm. (Make : Polycab/Suncab/Indoplast/Surya)	Nos	15	
125	House service bracket (Hook type) complete with 3 m long GI pipe 40mm dia medium grade with two Nos 90° bend, MS hooks/clamps and stay of guy.	Nos	1	
126	MCB DB 8 way SPN, 240 volt conform to IS-13032. (Make : Havells/Indo Asian/Legrand)	Nos	2	
127	MCB 16 A SP, 240 Volt breaking capacity 10 KA conform to IEC-60898(2002) and IS8828(1996). (Make : Havells/Indo Asian/Legrand)	Nos	6	
128	MCB 40 A DP, 240 Volt, breaking capacity 10 KA conform to IEC-60898(2002) and IS8828(1996). (Make : Havells/Indo Asian/Legrand)	Nos	2	
129	PVC insulated heavy duty 1100 volt grade cable with aluminium conductor twin core 10 Sqmm served with sheathing of PVC tape and overall PVC sheathed, flat or circular. Make - Plaza / Finolex / Havells	RM	180	
130	Insulation tape 19mm width, 25m long. (Make : Steel grip/Anchor)	Nos	4	
131	Exterior corner wall LED light 50W with 1.2 mtr 40mm dia pole with complete fittings etc	Nos	4	
132	No 8 x 19 CSK screw full threaded	Pkt	2	
133	No 8 x 32 CSK screw full threaded	Pkt	2	
134	No 8 x 50 CSK screw full threaded	No	2	
135	No 8 x 75 CSK screw full threaded	Pkt	2	
136	No 8 x 150 CSK screw full threaded	Pkt	2	
137	Geyser 25 Litre capacity with ISI marked incl bracket and all fitting accessories, first grade superior quality as per approved sample. (Make : Havells/ Philips/Usha/ Bajaj)	Nos	1	
138	PVC Connection 450mm long for cold & Hot	Nos	2	
Misc Items				
139	Mason thread (Gola)	Bdl	2	
140	Solvent cement for PVC pipe	Kgs	1	
141	Wood screw ½" (Pkt of 100)	Pkt	2	
142	Lime unslacked	Kg	3	
143	Slump Test kit	Nos	1	
144	Tile cutter machine. Make- Bosch/Hitachi/Stamenly	Nos	1	
145	Tile cutter blade. Make- Bosch/Hitachi/Stamenly	Nos	2	
146	Heat Gun flat nozzle adjustable 50 degree to 600 degree film removal power rating 1500-2000 W Make- Bosch/Stanley/Black Decker/Makita.	Nos	1	
147	Iron Cutter Blade 14". Make- Bosch/ Hitachi/ Stamenly	Nos	3	

Earthing Items.			
148	Electric Earthing Set complete with items as given below :- 148.1 GI Earthing plate 600mm x 600mm x 6mm thick with two Nos holes - 01 No. 148.2 GI pipe 20mm light grade for watering - 1.2 Mtr. Make : Tata/Jindal/Prakash 148.3 GI wire 4mm dia - 02 Kgs. 148.4 GI funnel with wire mesh suitable for 20mm dia GI pipe - 01 No 148.5 GI strip (32mm x 6mm x 4.00mtr) 148.6 Common salt - 50 Kgs. 148.7 FRP earthing pit with cover - 01 No. 148.8 GI nut bolt with washer of size 10mm dia 25mm long - 2 Nos. 148.9 Charcoal - 50 Kgs.	Nos	1
149	LC set with safe chemical 149.1 Zinc & Alloy Coated Safe Earthing Electrode (76.3mm dia) 2 mtr long with back frilled compound conductotile - 2 Bags CPRI approved 149.2 Air Terminal single pointed aluminium rod 12mm dia and 600mm. - 01 Set 149.3 GI pipe med graded 6 mtr long 40mm dia. Make TATA/JINDAL/BANSAL - 01 Nos 149.4 Insulator with screw - 10 Nos 149.5 Aluminium cable 70Sqmm multi standed single core. Make PLAZA/HAVELLS/ RALLISON ISI Marked - 10 RM. 149.6 Testing point terminal box made out of gun metal of phosphorus of size 75mmx75mmx5mm with screw. 149.7 Nut and bolts 2" x 1/2" - 08 Nos 149.8 Foundation Nuts & bolts 16mm x 400mm - 04 Nos. 149.9 Aluminium lugs 70Sqmm for connection of cable with nut & bolts of suitable size - 02 Nos. 149.10 Earthing pit FRP Incl Cover - 01 Nos. 149.11 PCC Brick size 230x100x75 mm - 15 Nos 149.12 Sand - 0.15 Cum 149.13 Stone Agg 20 mm - 0.25 Cum 149.14 Cement - 1 Bags	Set	1

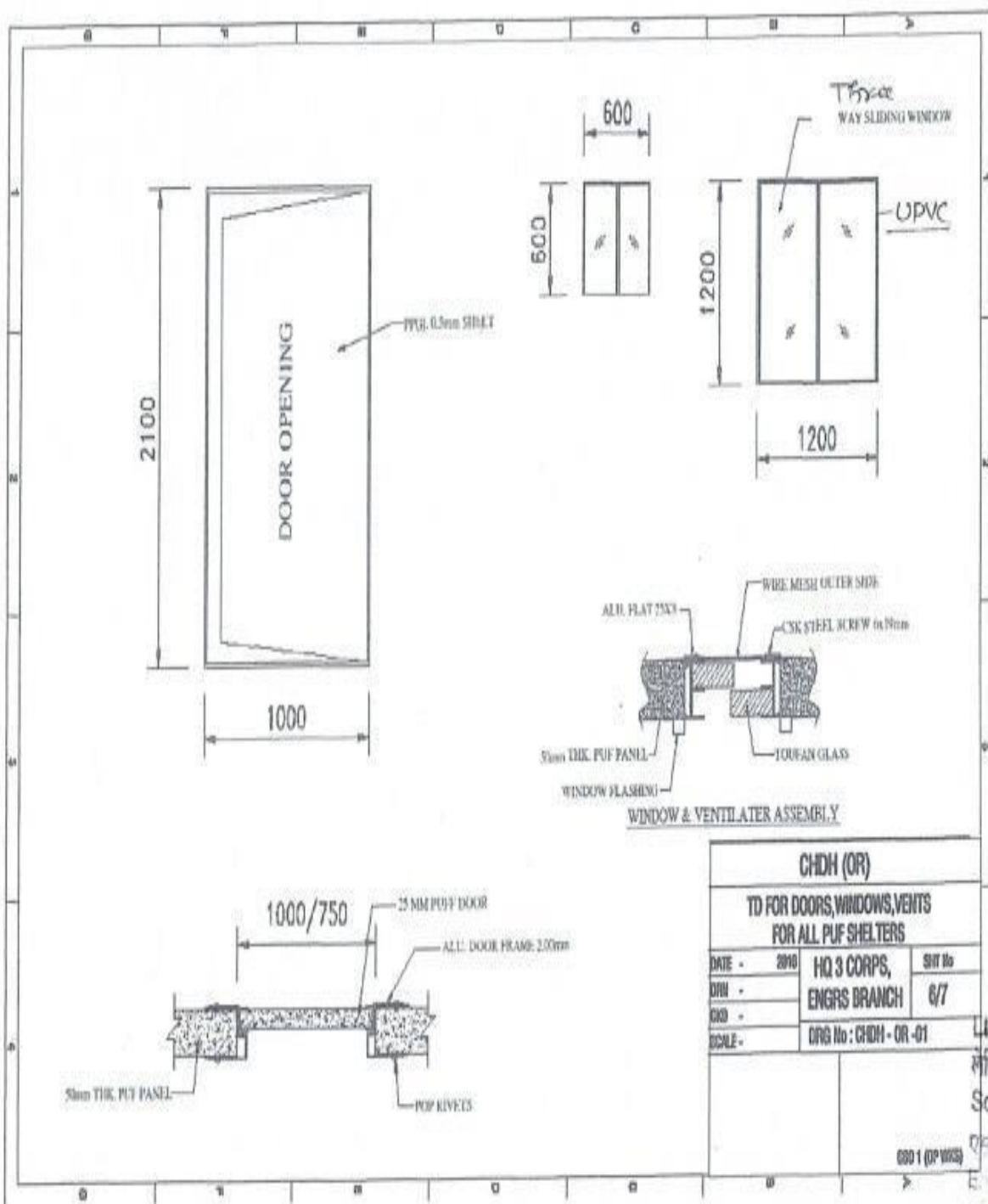


Li Col
 लिफ्टिनेंट कर्नल
 So1 (Engrs)
 एस० ओ० १ (अभियन्त)
 Engrs Branch
 अभियन्त शाखा
 HQ 3 CORPS
 मुख्यालय ३ क०
 Pin-808503

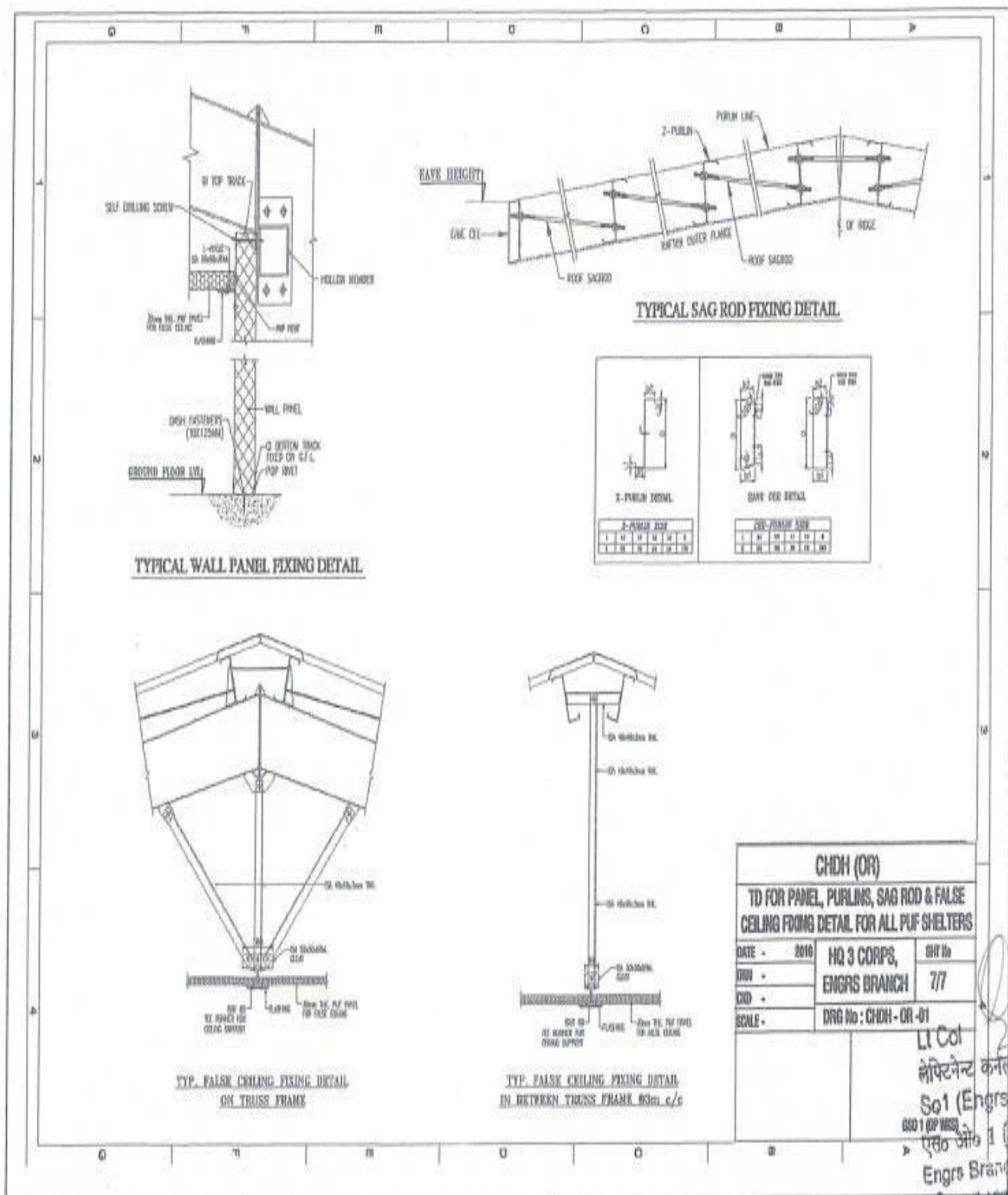
FT-01 610 RIDGE CAP	FT-02 180 OUTER CORNER	FT-03 100 INNER CORNER
FT-04 115 BOTTOM TRACK	FT-05 115 TOP TRACK	FT-06 040 CLOSURE TRIM - I
FT-07 180 WINDOW FLASHING	FT-08 170 DOOR FLASHING	FT-09 155 T CAP FLASHING

CHDH (OR)		
TRIM AND FLASHING DETAIL FOR PUF SHELTER		
DATE - 2010	HQ 3 CORPS,	SHT No
DRN -	ENGRS BRANCH	5/7
CND -	DRG No : CHDH - OR -01	
SCALE -		
		ESO 1 (DP INKS)

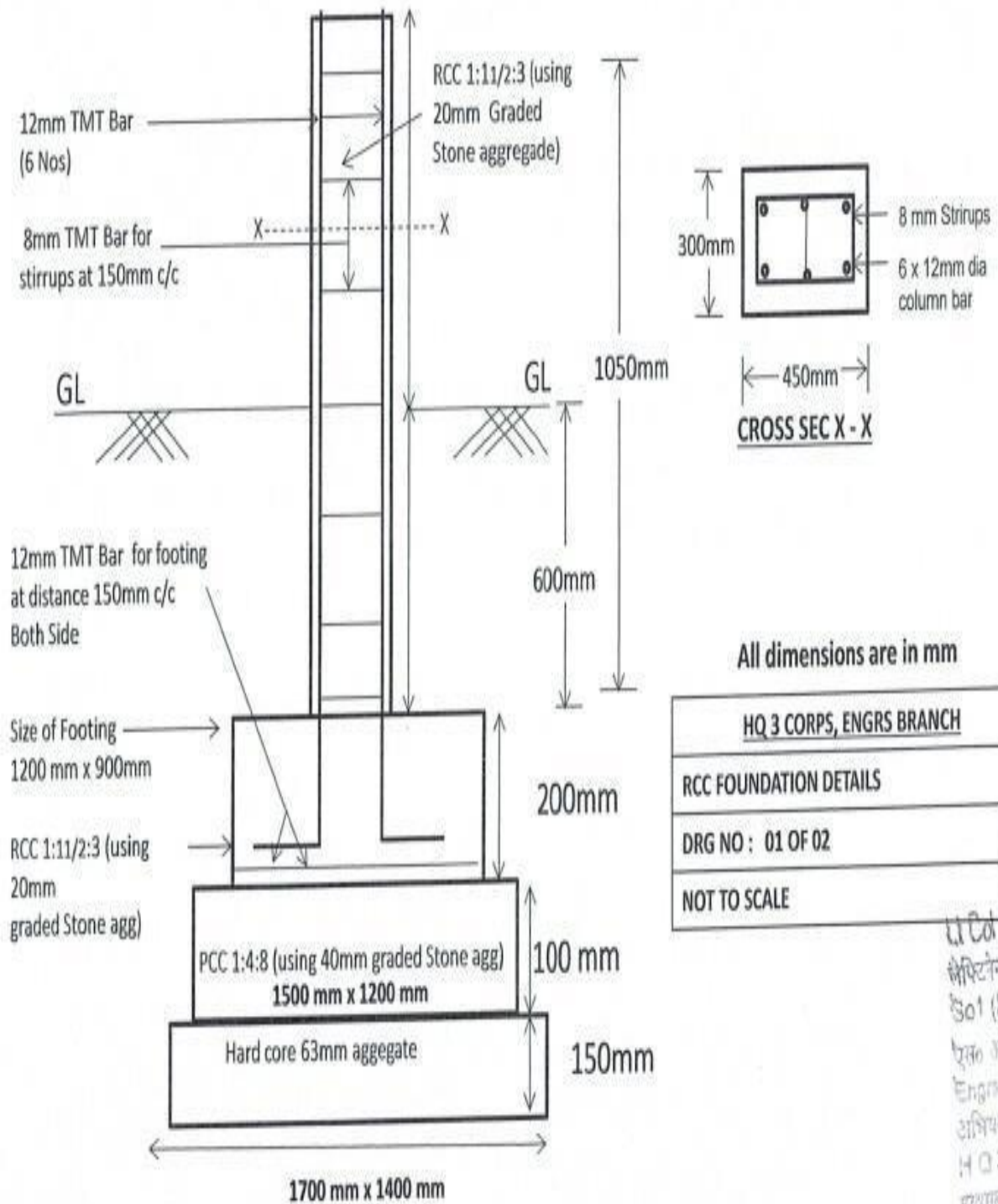
Lt Col
 लेफ्टिनेंट कर्नल
 S01 (Engrs)
 एंग्र्स ब्रान्च
 एंग्र्स ब्रान्च
 अतिरिक्त
 H Q 3 CORPS
 मुख्यालय 3 कोर
 Pin-908503



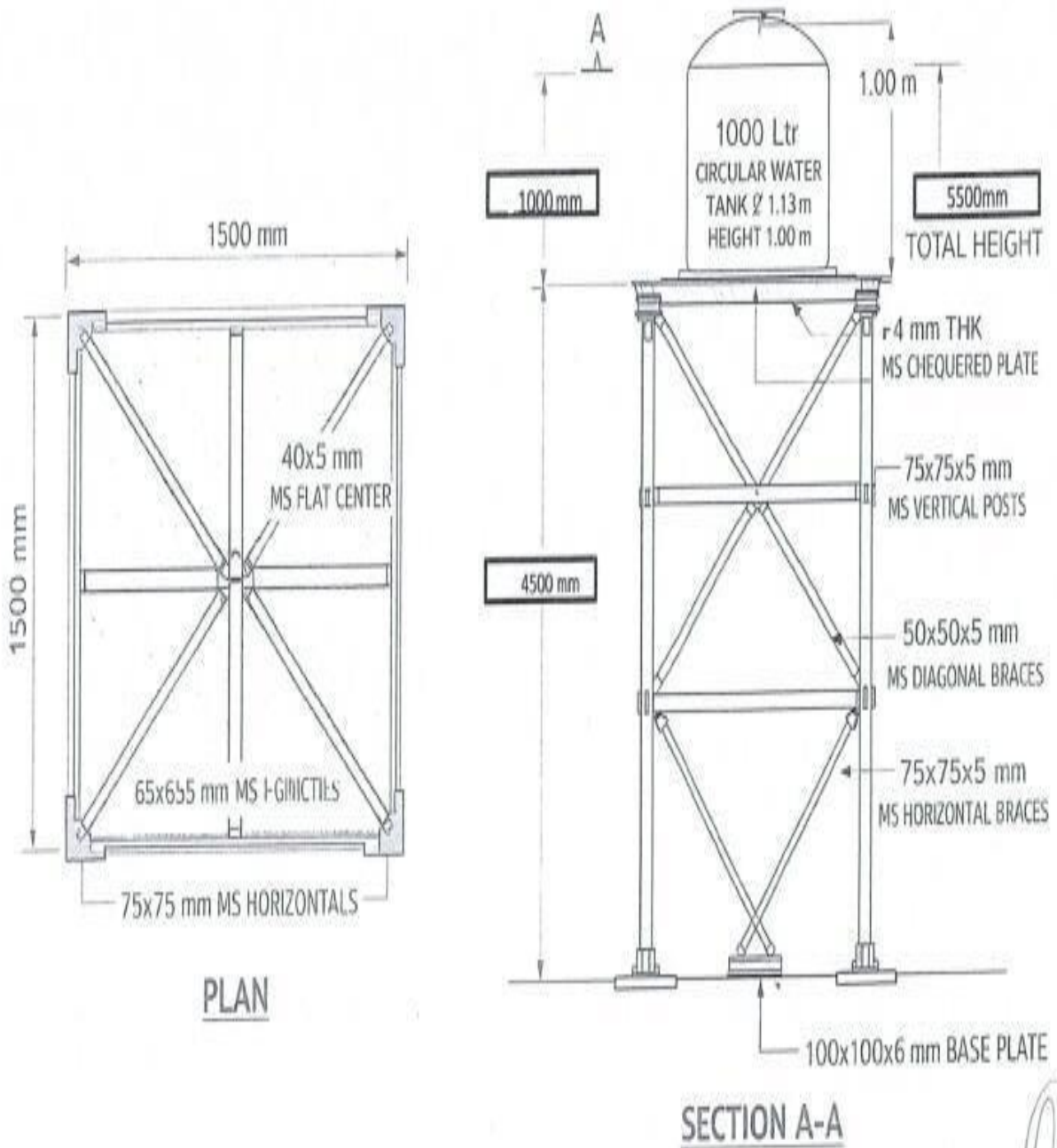
Lt Col
 सॉप्टवेयर कर्नल
 Sd1 (Engrs)
 एन को 3 (अभियन्ता)
 H.Q. - 30098
 मुख्यमन्त्र 3 कार
 Pin-908503



DETAILS OF RCC COLUMN WITH FOOTING



U.C. Col
 मेजर एन. के. सोलंकर
 Sot (Engrs)
 एस. ओ. 1 (अभियन्ता)
 Engrs Branch
 अभियन्ता कार्यालय
 HQ 3 Corps
 मुख्यालय 7 कोर
 गिन-8088509

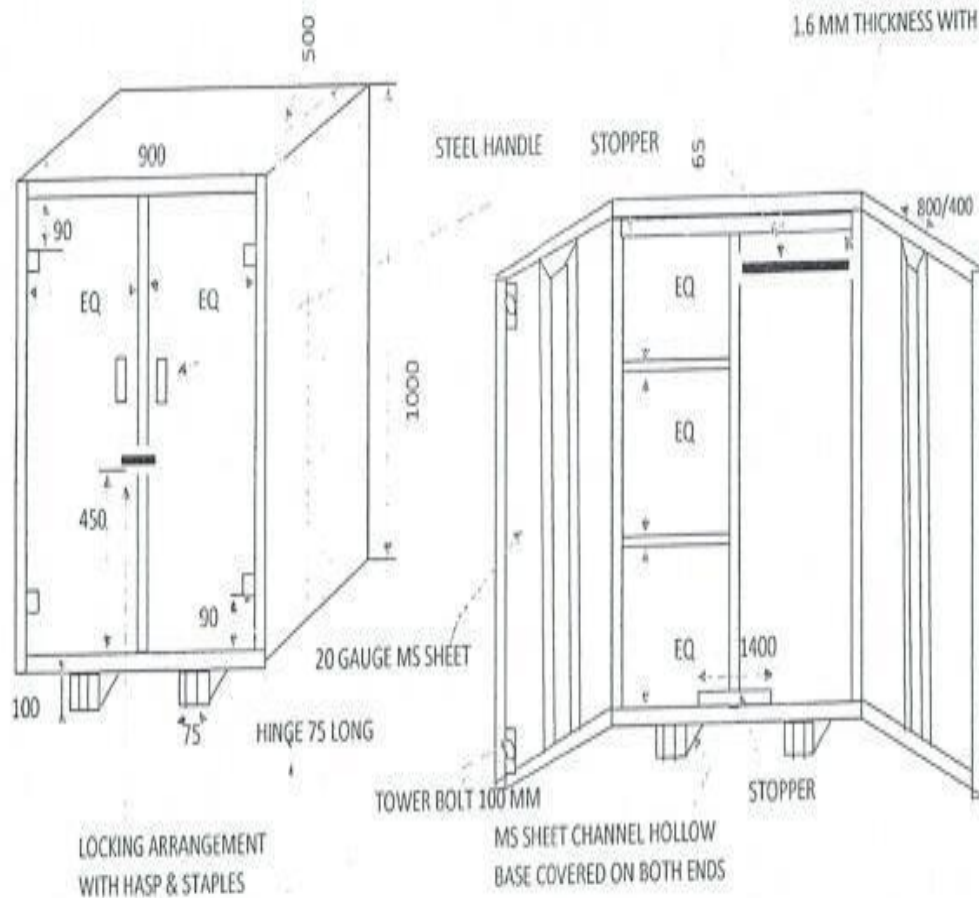


PLAN AND CROSS SECTION OF TANK STAGING

Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एसो ऑफो १ (अभिधन्ता)
Engrs Branch
अभिधन्ता शाखा
HQ 3 Corps
मुख्यालय ३ कोर
Pun-908500

BED SIDE LOCKER

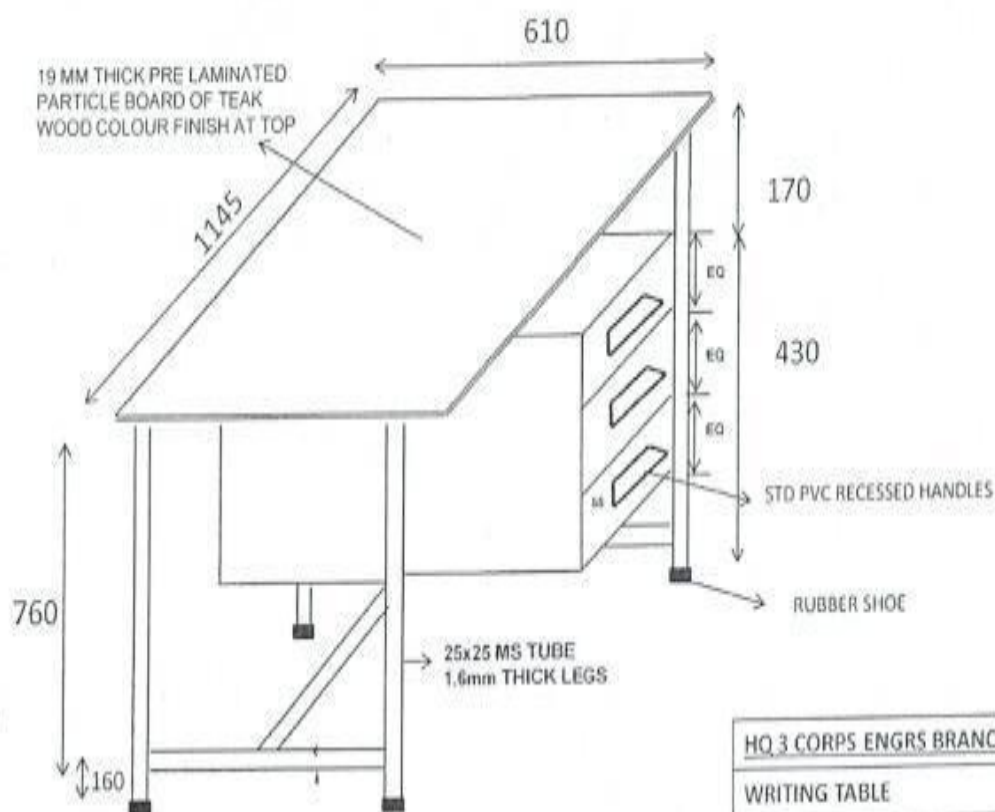
20MM STAINLESS STEEL CONDUIT PIPE
1.6 MM THICKNESS WITH SOCKET



NOTE : ALL DIMENSIONS ARE IN MM
AND DRAWING NOT TO SCALE

H Q 3 CORPS, ENGRS BRANCH
BED SIDE LOCKER
DRG NO : BSL 1/1
GSO 1 (OP WKS)

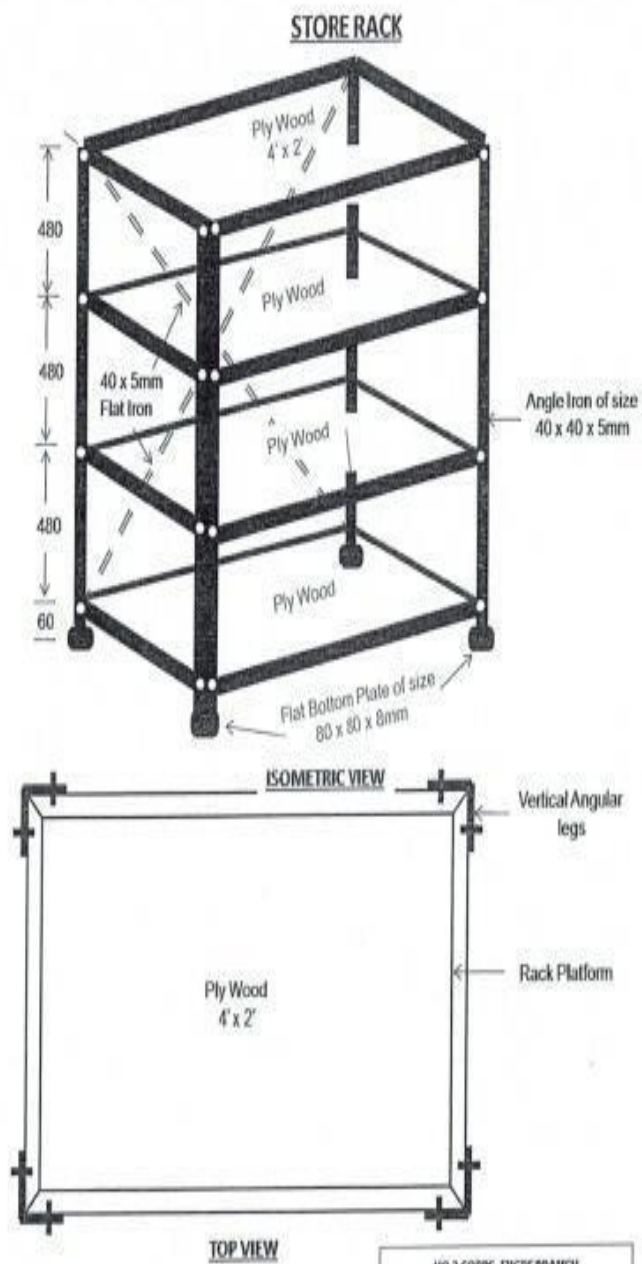
1. Col
कपिल
BSL (Engrs)
अभियन्ता
10/10/2023

WRITING TABLE

NOTE : ALL DIMENSIONS ARE IN MM AND NOT TO SCALE

HQ 3 CORPS ENGRS BRANCH
WRITING TABLE
DRG NO : WT 1/1
GS01 (OP WKS)

Lt Col
 लेफ्टिनेन्ट कर्नल
 So1 (Engrs)
 एसो ओ 1 (अभियन्ता)
 Engrs Branch
 अभियन्ता शाखा
 H Q 3 Corps
 मुख्यालय 3 कोर
 Pin-908503

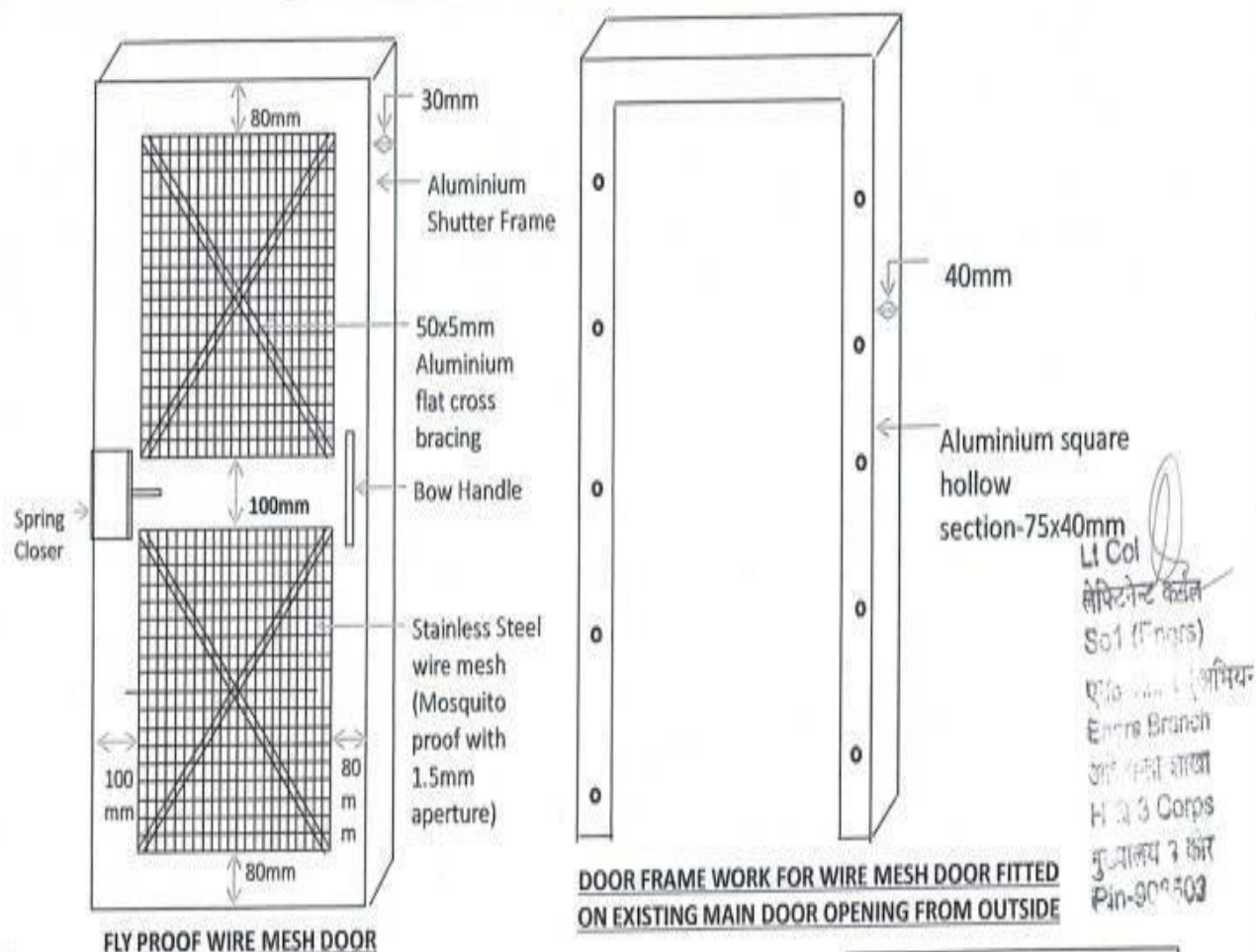


HQ 3 CORPS, ENGRS BRANCH
STORE RACK
DWG NO : SR - 1/1
GSO 1(OP WKS)

NOTE > DRG NOT TO SCALE (ALL DIMENSIONS ARE IN MM)

Lt Col
 लेफ्टिनेंट कर्नल
 So1 (Engineering)
 एंगो डि. 1 (अभियन्ता)
 Engrs Branch
 उप निरीक्षक
 H Q
 मुख्यालय
 Pin-800002

FLY PROOF WIRE MESH DOOR DETAILS



Note :-

1. Drg not to scale .

HQ 3 CORPS, ENGRS BRANCH
FLY PROOF DOOR FOR PUF SHELTER
DRG NO : FPD 1/1
GSO1 (OP WKS)

SPECIAL CONDITIONS

1. **INSPECTION OF SITE.** It will be ensured by the tenderer that the site visit is undertaken by him/her before placing the bid. No representation in this regard shall be entertained.
2. All precautions shall be taken by the contractor in demolition/dismantling of items. Any damage to existing structure/fittings during course of demolition/dismantling shall be made good by the contractor at his own cost and without any extra cost to the Govt.
3. The location of plant, labour camp, contractor's site office & lab and routes to follow to reach the plant and site shall be as approved by local mil authorities. Contractor shall not be allowed to follow any other route. Changes may, however, be made in the route on any later date. Nothing extra shall be paid to the contractor in this regard.
4. All the works pertaining to this contract shall be supervised by **110 Engineer Regiment.**
5. The equipments and T&P to be deployed by Contractor shall be in good working condition.
6. Where reinforcement details/thickness of RCC members such as slab, beams, lintels etc are not shown on the drawings, the same shall be supplied by the EPO on the basis of provisions in IS-456- 2000 and IS-13920, the same shall be provided and incorporated in the works by the contractor and no extra payment will be admissible on this account.
7. **CONDITIONS OF WORKING IN RESTRICTED AREA.** The entire work under this contract lies in "RESTRICTED AREA". Restriction for entry to work site and working conditions in Restricted Area shall be as under:-
9. **ENTRY AND EXIT.** The contractor/his agent(s)/ representative(s)/ workmen etc., and his materials carts, trucks or other means of transport, etc., will be allowed to enter through and leave from only such gate or gates and at such times as the EPO or authorities in charge of the restricted area may at their sole discretion permit to be used. The contractor's authorised representative is required to be present at the places of entry and exit for the purpose of identifying his carts, trucks, etc, to the personnel- in-charge of the security of the restricted area.
10. **IDENTITY CARDS OR PASSES: -**
 - 10.1. The contractor, his agents and representatives **are required individually to be in possession of an identity card or pass duly issued by station authorities or EPO as per latest instruction applicable at station.** The identity card or pass will be examined by the security staff at the time of entry into or exit from the restricted area and also any time or number of times inside restricted area.

10.2. **IDENTITY OF WORKMEN.** Every workman shall be in possession of an Identity Card. The identity cards shall be issued after a thorough vestigation of the antecedents of the labourers, by contractor and attested by the officer-in-Charge of the gen area of wk oncerned in accordance with the standing rules and regulations of the unit.

10.3. The contractor shall be responsible for conduct of his workmen, agents or representatives.

11. Workers/supervisor/contractor and their agents are not allowed to carry smart mobile phone having features like camera, net connection and related apps to the work site. Smart phone are to be deposited with security personnel's of the unit/station. If any worker/supervisor/contractor and their agent found with smart phones at work site then said phones will be seized by the concerned unit/station security authorities for further action.

11.1. **SEARCH.** Thorough search of all persons and transport shall be carried out at each gate and for as many times as gate is used for entry or exit and may also be carried out any time or any number of times at the work site within the restricted area.

11.2. **WORKING HOURS.** The units controlling restricted area, usually work during six days in a week and remain closed on the 7th day. The working hours available to contractor' labor/staff shall however accordingly get reduced because of the time taken in security checks observed at the time of entry and exit and during working hours. The exact working hours, working days and non working days observed for these restricted area(s), where works are to be carried out shall be deemed to have been ascertained by contractor before submitting his tender. The tenderers attention is invited to the fact that total number of working hours for a unit are prescribed in regulations and that they cannot be increased by the EPO. The definition of "Working Day" as given under condition 1 (t) of IAFW-2249 does not apply in case where the works are carried out in restricted area.

11.3. **WORK ON HOLIDAYS.** The contractor shall not carry out any work on gazette holidays, weekly holidays and other non-working days except when he is expressly authorized in writing to do so by the EPO. **The CO may declare any day as non- working day due to exigencies of service.**

11.4. **ACCESS TO RESTRICTED AREA AFTER COMPLETION.** After the works are completed and surplus stores etc. removed, the contractor, his agents, representatives or workmen etc., will not be allowed any access to the restricted area except for attending to any rectification of defects pointed out to him by the Commanding Officer, 110 Engr Regt, Engineer-in-Charge.

11.5. **FIRE PRECAUTIONS:-**

11.5.1. The contractor, his agents, representatives, workmen etc, shall strictly observe the orders pertaining to fire precautions prevailing within the restricted area.

11.5.2. Motor transport vehicle, if allowed by the authorities to enter the restricted area must be fitted with serviceable fire extinguishers.

12. **SECURITY OF CLASSIFIED DOCUMENTS.** Contractor's special attention is drawn to condition 2-A & 3 of General Conditions of Contracts (IAFW-2249). The contractor shall not communicate any classified information regarding the works either to Sub- contractors or others without prior approval of the Commanding Officer, 110 Engr Regt, Engineer-in- Charge. The contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of the work and he shall return all documents on completion of the work or earlier or termination of contract. The contractor shall along with final bill attach a receipt from the EPO in respect of his having returned the classified documents as per condition 3 of General Conditions of contracts (IAFW-2249).

13. **OFFICIAL SECRET ACT.** The contractor shall be bound by the Indian Official secret Act 1923, particularly section 5 thereof, all as per condition 2A of IAFW-2249.

14. **EMPLOYMENT OF PERSONNEL.** Contractor shall employ only Indian Nationals as his representatives, servants and workmen after verifying their antecedents and loyalty. **He shall ensure that no person of doubtful antecedents and nationality is, in anyway, associated with the work.** If for the reasons of technical collaboration or other considerations the employment of foreign national(s) is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of the tender. As a proof that the contractor has employed only Indian Nationals, he shall render a certificate to EPO within one month from the date of acceptance of tender to this effect. In case the Accepting Officer desires, contractor will have the police verification done of personnel employed by him.

15. **AVAILABILITY OF LAND FOR STORAGE OF MATERIALS, TEMPORARY WORKSHOP AND OFFICES ETC**

15.1. **ALLOTMENT OF LAND FOR STORAGE OF MATERIALS INCLUDING TEMPORARY WORKSHOP AND OFFICE.** Refer condition 24 of IAFW-2249. The contractor shall be permitted to store his materials including erecting temporary sheds at the area of land marked on site plan and he shall pay a license fee of Rs 1/- per year or part thereof in respect of each and every separate area of land allotted to him.

15.2. **ALLOTMENT OF LAND FOR FABRICATING WORKSHOPS ETC.** Refer condition 24 of IAFW-2249. The contractor shall be permitted to erect his temporary workshops and the like at the area of land marked for the purpose on the site plan and he shall pay a license fee of Rs. 1/- (Rupees one only) per year or part thereof in respect of each and every separate area of land allotted to him.

16. **APPROACHES.** The contractors shall make arrangements for and provide at his own cost all temporary approaches, if required to the site(s), after obtaining approval in writing of the Engineer-in-Charge/EPO to the layout of such approaches.

17. **LOCATION OF BUILDINGS AND WORKS.** There may be some changes in location/sitting of building shown in site (layout) plan (s) to suit local conditions and/or departmental requirements. The contractor shall have no claim what-so-ever consequent

to such change in the location/sitting of works.

18. MINIMUM WAGES PAYABLE.

18.1. Refer condition 58 of IAFW-2249. Govt. of India Min of Labour (Deptt of labour and Employment) vide notification dated have fixed minimum rates of wages as up to dated from time to time for various categories of employees under the Minimum Wages Act 1948. The contractor shall not pay wages lower than minimum wages including allowances as fixed under said notification as updated till the date of receipt of tender. Where in any area the minimum rates of wages are also fixed by state Government for employees in employment in the construction/maintenance of buildings/roads/runways, the labour wages as fixed vide above said updated notification of Ministry of labour or respective state Government whichever is higher in respect of these areas shall be deemed minimum rates of wages payable under this contract and the contractor shall not pay lower than this minimum rates of wages.

18.2. Contractor's attention is also drawn, amongst other things to the "explanations" to the schedule of minimum wages referred to above.

18.3 The fair wages referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages, referred to above as up to dated from time to time.

18.4 Sch. of minimum wages has not been enclosed along with tender documents. However, contractor shall be deemed to have verified the minimum wages payable as on the last due date of receipt of tender.

18.5. The contractor shall have no claim whatsoever, if on account of local factor and/or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

18.6. Wages/salary of employed labourers and supervisor/deployed staff shall be paid by the contractor through Aadhar Linked bank account of individual labour and deployed staff. Relevant details of payment made by the contractor through bank EPO shall be produced for verification of principal employer (EPO).

19. TAXES.

19.1. Tendered rates are inclusive of recently rolled out

19.2. Goods & Services Tax (GST) made applicable from 01 July 2017

19.3. Labour welfare cess.

19.4. Other levies, royalties & duties.

20. EQUIPMENT AND MATERIALS:- All equipment and materials to be incorporated in the work shall be new and of the best of their kind conforming to the latest IS or in their absence to British or other relevant standard specifications.

21. TESTING OF MATERIAL. The contractor shall be responsible for various tests to be carried out as listed in Appx 'A' in the site laboratory which will be set up by the contractor for testing of material at his own cost and arrangement and for that, they shall employ a competent technical representative as approved by EPO. All such tests shall be carried out in presence of Engineer-in-Charge. The cost of labour & materials for testing

shall be borne by contractor. The quantity of items & work mentioned in this supply order may be increased/decreased due to change in scope of work on the direction of the Commanding Officer, 110 Engineer Regiment. The Commanding Officer reserves the right to delete any item based on the assessed cost.

22. Joint Receipt Inspection (JRI).

22.1. JRI of stores shall be carried out at FOR/delivery location between the representatives of the firm and a board of officers (BOO) detailed by the consignee's formation within 96 hours of the delivery. The JRI shall consist of:-

22.1.1. Quantitative checking to verify that the quantities of the delivered items correspond to the quantities defined in the supply order and the invoices. In case of ambiguity, handbook for various established norms will be followed in consultation with CO, 110 Engineer Regiment.

22.1.2. Complete functional checking of the stores/equipment as per specifications in the supply order.

23. Upon completion of JRI, the proceedings shall be signed by the BOO and representatives of the firm. In case the representative of the firm is not present, the board proceedings shall be signed by the BOO alone and the same shall be binding on the firm. Deficiencies in Quantity, Quality or Defects shall be recorded in the JRI proceedings and the firm informed in writing to replace/make up the deficiencies/defects.

24. In case of rejection of goods in full or in part during JRI, the items so rejected shall be removed from the delivery point by the firm at their own cost within 96 hours. The rejected stores shall be replaced by the firm at delivery point free of cost within fifteen days of receipt of intimation from the consignee. On failure to do so, the consignee will be at liberty to purchase the rejected stores at the firms risk and cost.

25. **Termination of Supply Order.** In the event, items offered by you repeatedly fail in inspection for more than two times or you are not able to rectify the defect at the first instance itself and offer items not conforming to the quality standards mentioned in the Supply Order, the Supply Order offered to you for those items shall be terminated/cancelled instantly.

Delivery.

26. **Freight.** Packaging and freight will be sole responsibility of the supplier. Damage/losses in transit shall be the responsibility of the supplier upto the construction site (s) specified in clause 3 of the Supply Order. The stores will be securely packed and transported without any additional cost to the Govt. to avoid loss or damage during transit so as to reach the consignee in perfect serviceable and fit for use condition. The supplier shall remain responsible for adequacy of packages during transit. The packages shall be marked with the Supply Order and other necessary details for easy identification at the consignee's end. The supplier will be responsible for unloading of stores at FOR and to ensure that the stores are delivered in sound condition without any damage.

27. **Marking.** All the major components will be marked with the following details using suitable stencil or by screen printing:-

27.1. Manufacture/Firm's Name.

- 27.2. Part No (if any).
- 27.3. Supply Order
- 27.4. Consignee.

28. **Risk and Cost Purchase.** In the event of breach of contract on any account or no supply of materials or delay in supply of work, Commanding Officer, 110 Engineer Regiment shall have the right to cancel the supply order and purchase the stores at your risk and cost. Additional expenditure incurred shall be recovered from your firm.

29. **Quality Assurance.** The materials supplied should conform to the ISI standards and specifications as mentioned in Supply Order. An Assurance certificate as per specifications will be produced by the supplier for the supplied quantity of the materials.

30. **Discrepancies.** If there are varying or conflicting provisions made in any of the documents forming part of the Supply Order, Commanding Officer, 110 Engineer Regiment shall be the sole deciding authority with regard to the intention and his decision shall be final and binding. If any item not mentioned in this SO is required for usage of the stores supplied as per good engineering practices, the same will be supplied by the contractor and is deemed to be included in the rate quoted by the contractor.

31. **BLASTING.** Blasting in any form is prohibited. Unless specifically permitted through written approval of competent authority.

32. **Loss Or Damage On Account Of Enemy Action.** Applicable for works in operational /border Area.

33. **EMPLOYMENT OF PERSONNEL.**

33.1. The contractor shall employ only Indian National as his representative, servants and workmen and verify their antecedents and loyalty before employing them for work. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with the works. If for reasons of technical collaboration or other consideration the employment of foreign national is unavoidable, the contractor shall furnish the particulars to this effect to the Accepting Officer at the time of submission of the tender. As proof that the contractor has employed only Indian Nationals he shall render a certificate to the EPO within one month from the date of acceptance of the tender to this effect. In case the EPO desires, contractor will have the police verification done of the personnel employed by him. Contractor's attention is also drawn to condition 25 of IAFW-2249 in this connection.

33.2. The Commanding Officer, 110 Engineer Regiment shall have full powers and without giving any reason, to require the contractor immediately to cease to employ in connection with this contract, any Agent or servant or employee whose continued employment is, in his opinion, undesirable.

33.3. **Contractor's Supervision:-** Refer Condition 25 of IAFW-2249 along with latest amendments. Contractor shall employ Engineering staff to supervise the execution of work/contract for the value of contract shall be as mentioned below:-

Contractor has to employ at least, one degree holder from a Government Recognized Institution or equivalent / final or direct final pass of sub - division II of the Institution of Surveyors (India) with at least 4 years practical experience of work.

OR

Contractor has to employ at least, two Diploma holder (one civil + One electrical) in Engineering from a Government recognized institution with at least 4 years practical experience of works.

34. **T&P, Machinery & Transport (Refer Condition 15, 16, & 34 of IAFW -2249):-** For smooth execution of work the contractor shall engage adequate quantities of T&P, machinery & transport to complete the work. In case of any discrepancy the decision of the CO 110 Engineer Regiment will be final to mobilize the required plant. Contractor will be liable to deploy at site any other tool or machine required as directed by Engineer -in-charge to execute specified nature of work or as per manufacturer's instructions.

35. Please note that you will be required to strictly ensure engagement of engineers / deployment of T&P, Machinery & transport as stipulated in the contracts. Inadequate engagement of engineers /deployment of T&P Machinery and Transport as per contract condition shall be considered as serious lapse attracting ban /removal / downgrading / debarment of the firm / Company. The above provision shall be deemed to be included in the quoted rate of contractor and nothing shall be admissible.

36. **Leakage/Seepage.** The contractor shall be liable for seepage/leakage in the bldg. for a pd of 10 yrs from the date of issue of completion for wk and shall take over repair at his own cost within 30 days of receiving of complaints through emails, letters from EPO. All necessary items like DPC designed to arrest the seepage should be used while constr of plinth and roof or as directed by Engineer -in Charge.

37. **Compensation.** Contractor/Firm fails to complete the work within PDC, 0.5% of contract value per week maximum to 5% of Contract value will be recovered from the contractor during part/final payment to the Contractor/Firm. Thereafter, Accepting Officer reserves the right to cancel the contract.

38. The Vendor should be in possession of last 3 years experience certificate of contracts and works executed with other offices within the Arunachal Pradesh. Vendor should submit the confirmation on **Awareness about the FOR location and construction site area** for the acceptance of the contract.

Sd/- x xx xxx
(Rishi Katiyar)
Col
Commanding Officer

